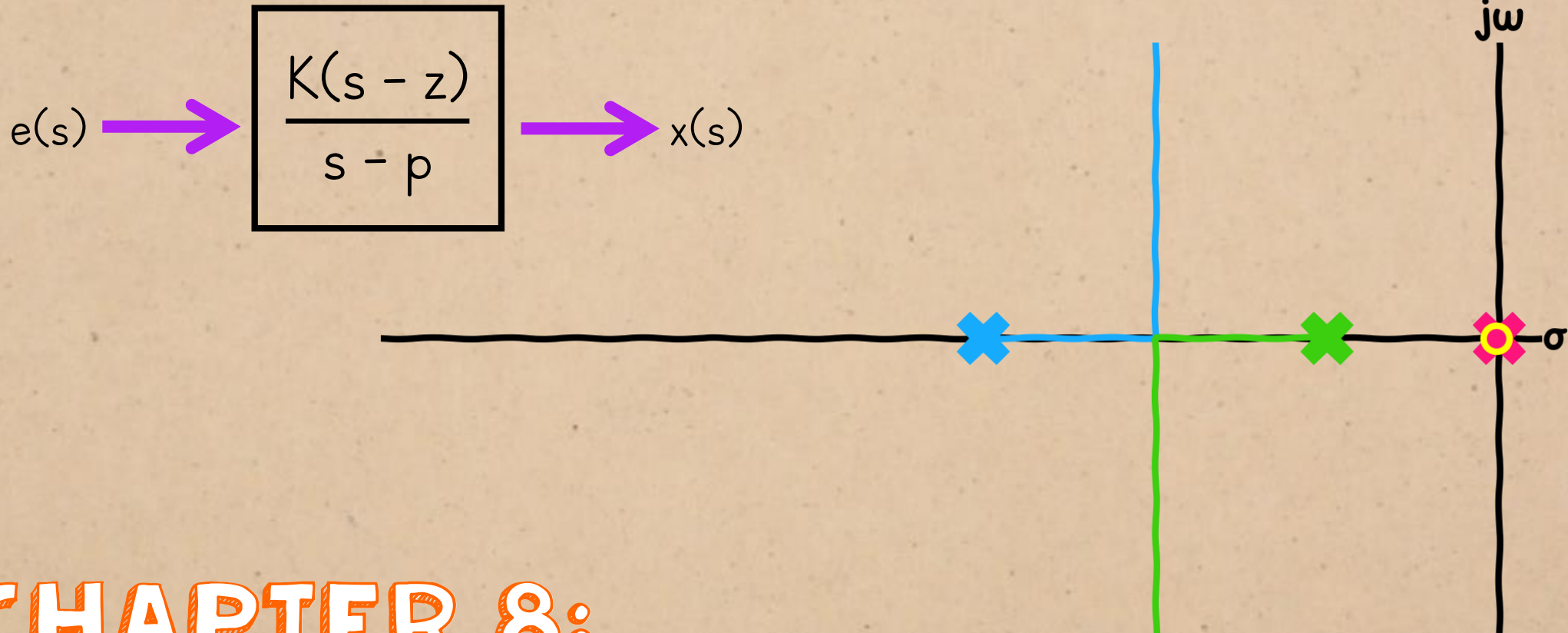




CHAPTER 8:

COMPENSATION

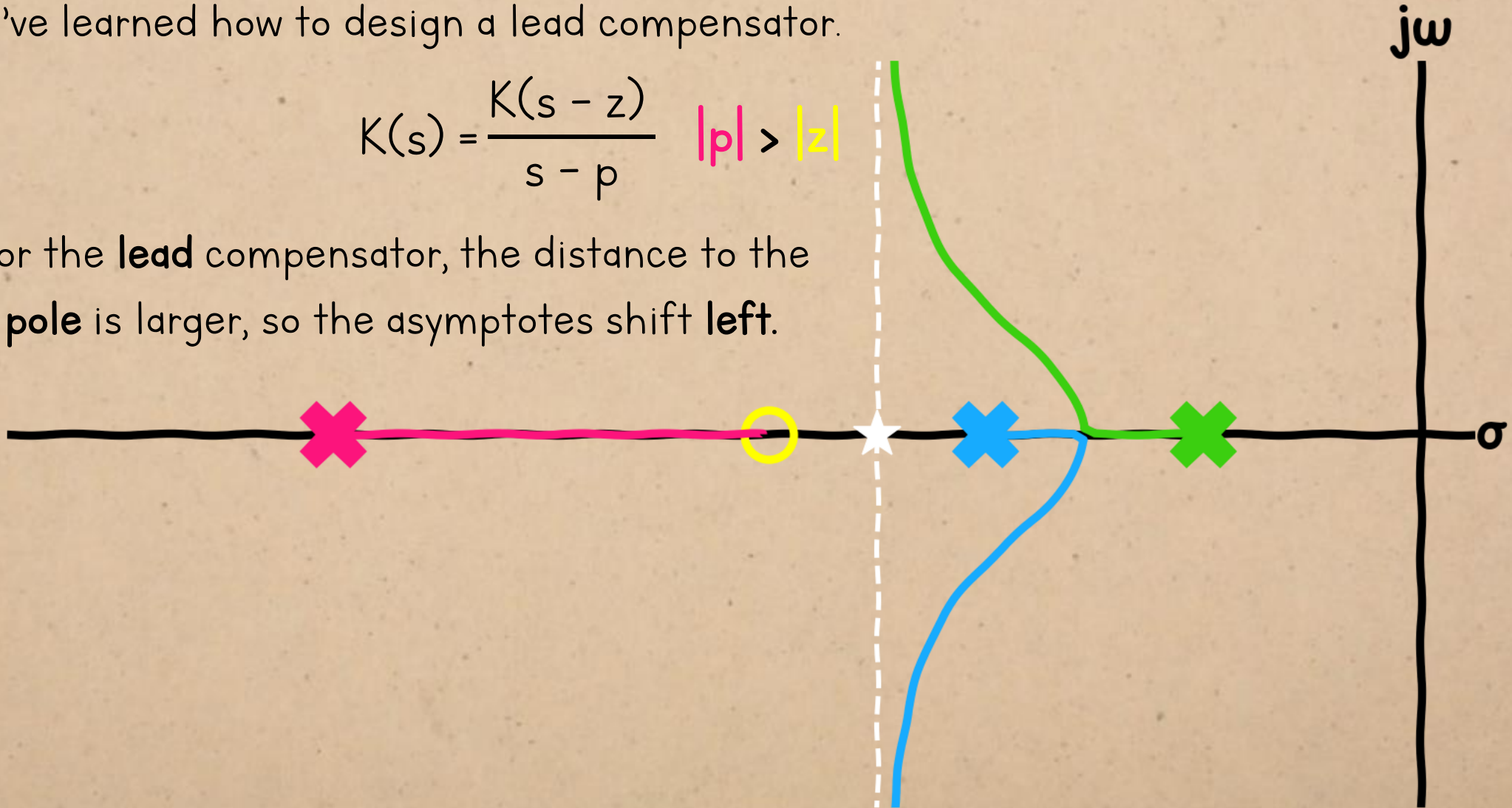
LAG COMPENSATOR DESIGN I

LAG COMPENSATOR

We've learned how to design a lead compensator.

$$K(s) = \frac{K(s - z)}{s - p} \quad |p| > |z|$$

For the **lead** compensator, the distance to the **pole** is larger, so the asymptotes shift **left**.

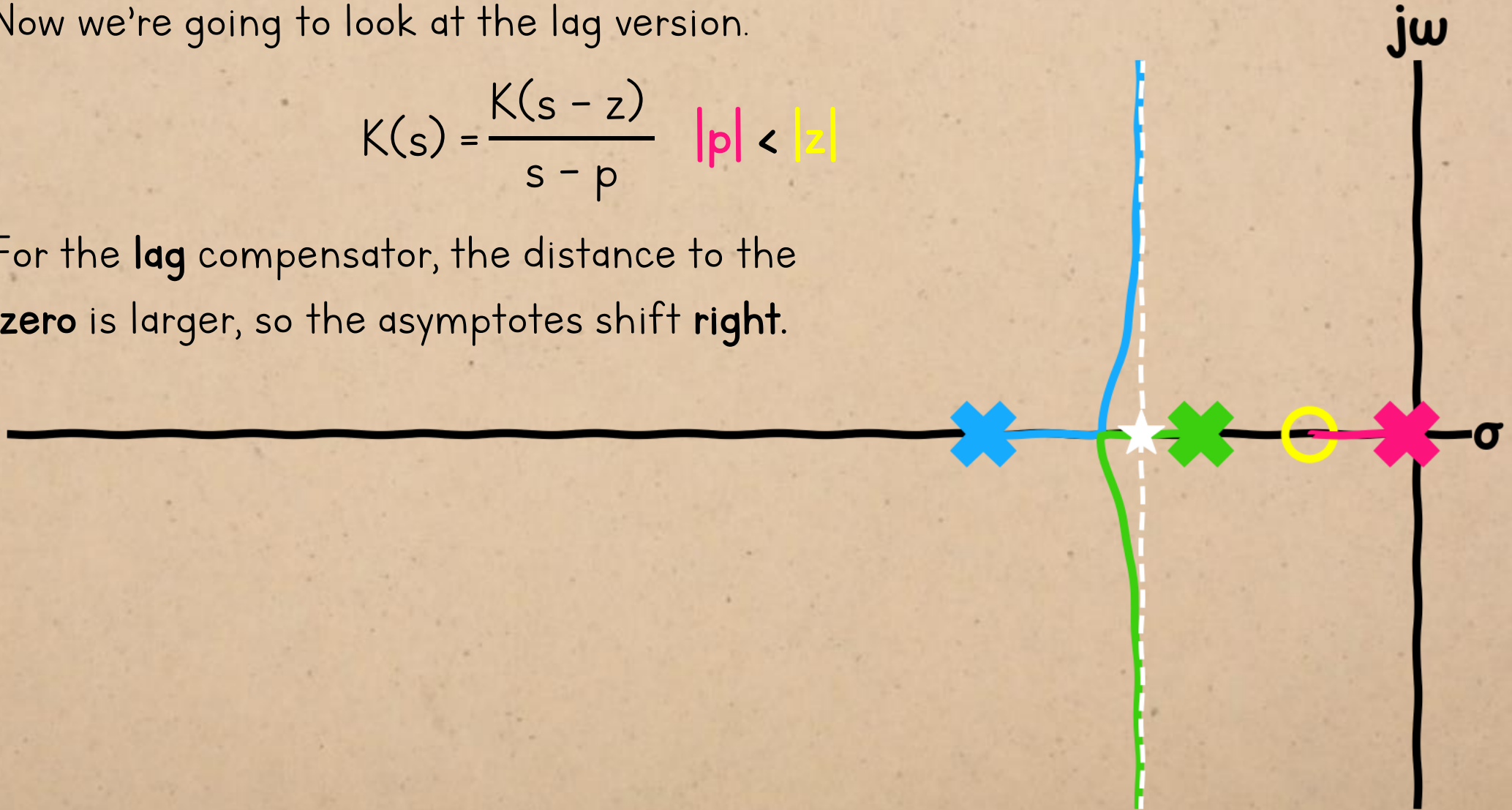


LAG COMPENSATOR

Now we're going to look at the lag version.

$$K(s) = \frac{K(s - z)}{s - p} \quad |p| < |z|$$

For the **lag** compensator, the distance to the **zero** is larger, so the asymptotes shift **right**.

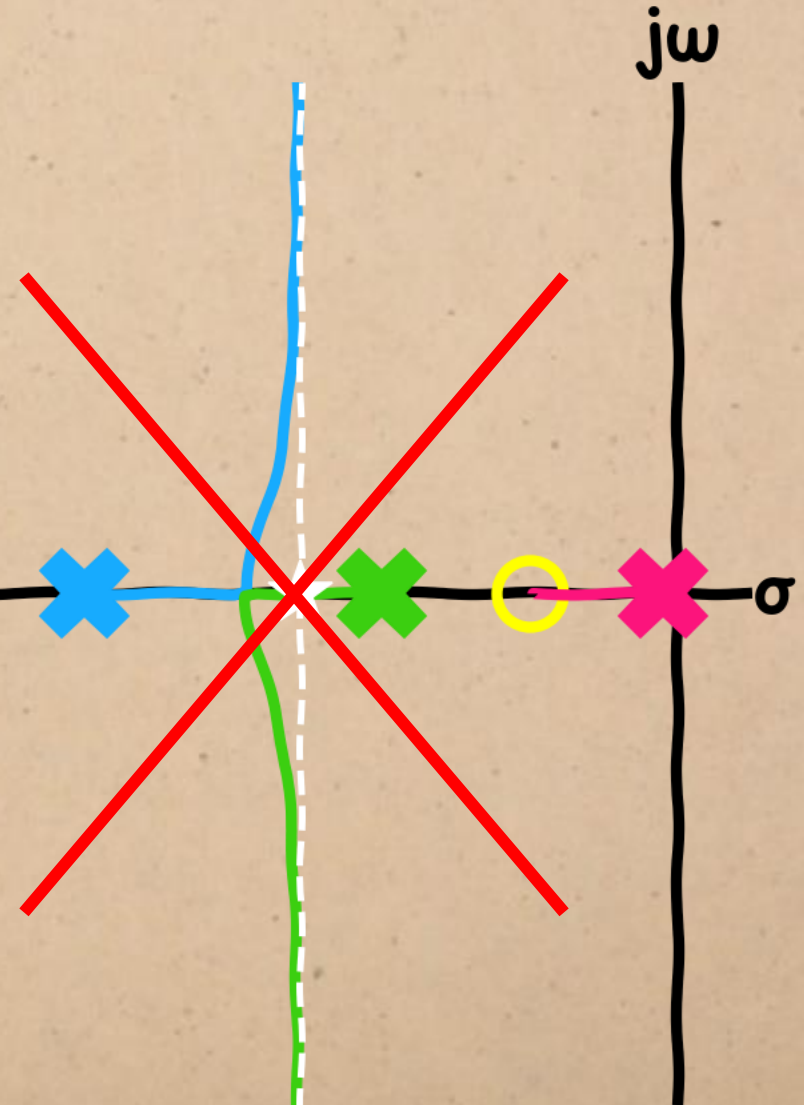


LAG COMPENSATOR

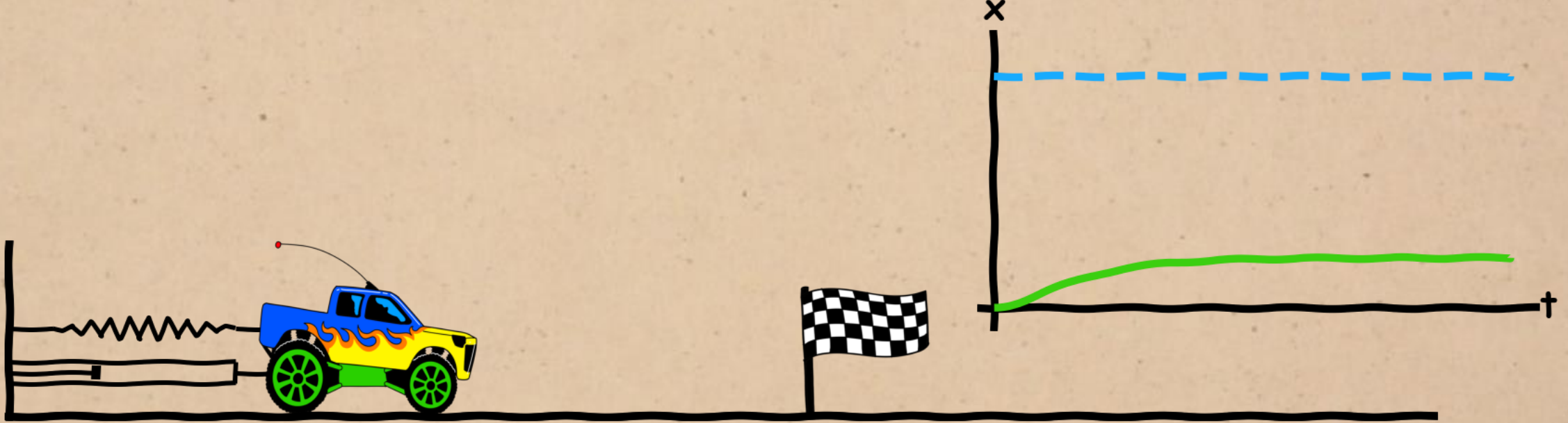
$$K(s) = \frac{K(s - z)}{s - p} \quad |p| < |z|$$

For the **lag** compensator, the distance to the **zero** is larger, ~~so the asymptotes shift right.~~

When we design the lag compensator, we want to minimize this rightward shift and change the shape of the root locus as little as possible.



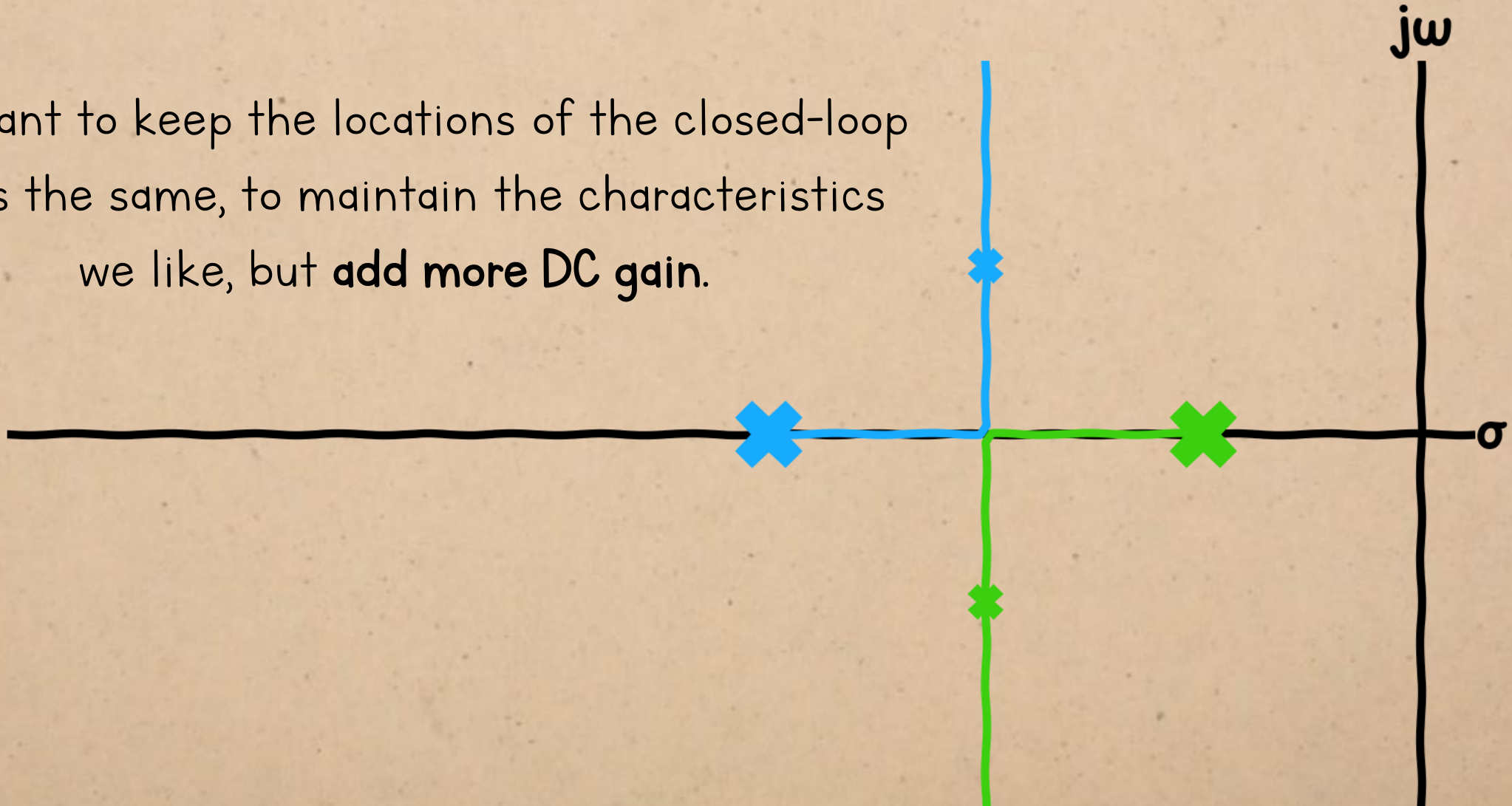
LAG COMPENSATOR



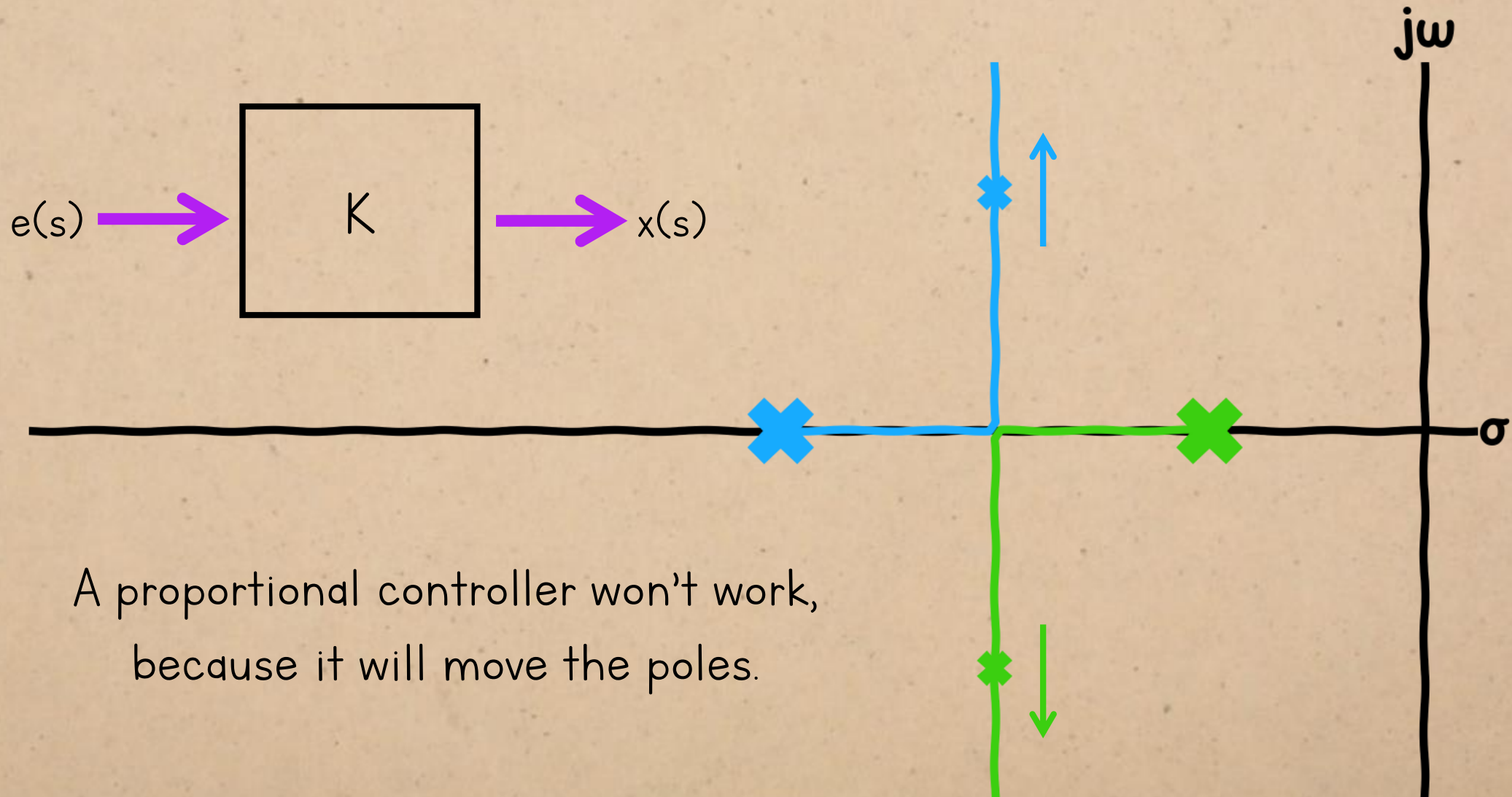
A lag compensator is useful when you have a system with an acceptable **transient response** but it doesn't have sufficient **steady-state accuracy**.

LAG COMPENSATOR

We want to keep the locations of the closed-loop poles the same, to maintain the characteristics we like, but **add more DC gain**.

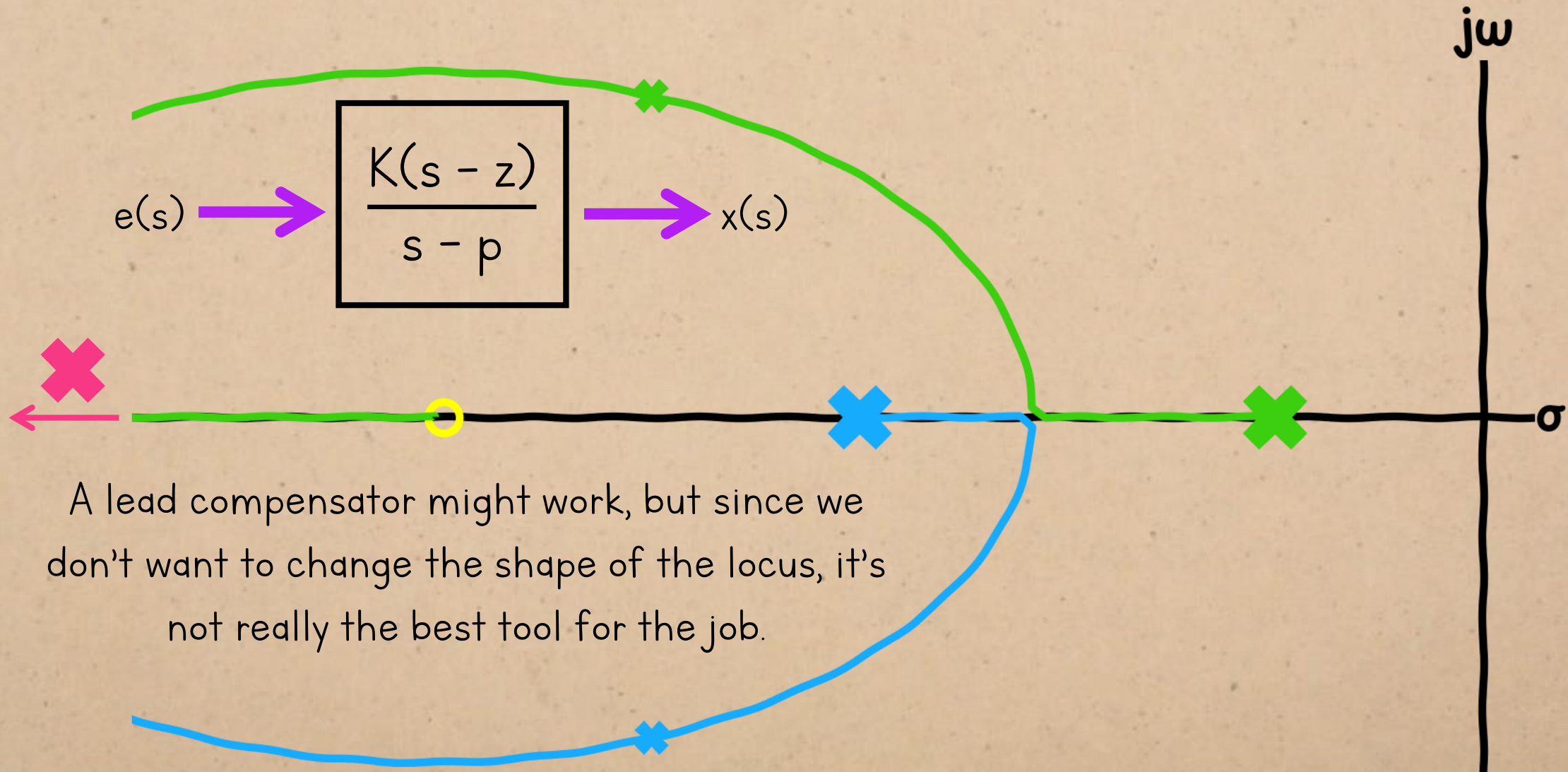


LAG COMPENSATOR



A proportional controller won't work,
because it will move the poles.

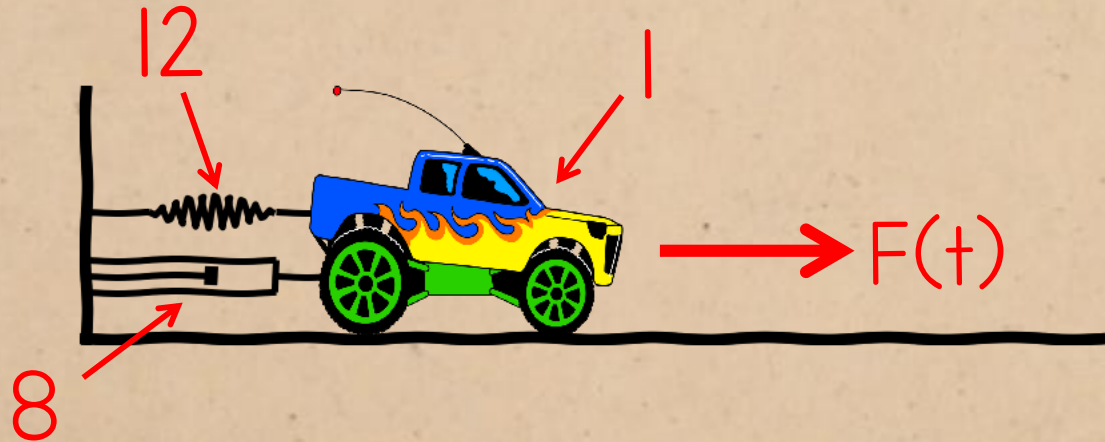
LAG COMPENSATOR



DESIGN SPECIFICATIONS

Considering the system:

$$\ddot{x}(t) = -8\dot{x}(t) - 12x(t) + 12F(t)$$



design a controller to meet the following requirements:

1. Position error $< 10\%$
2. % overshoot $< 1\%$
3. Settling time to 2% < 1 second

SOME MATHS

damping ratio:

The % overshoot is related to the damping ratio by

$$\% \text{ overshoot} = 100e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$$

$$\text{So, we need } e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \leq 0.01$$

$$\therefore -\frac{\zeta\pi}{\sqrt{1-\zeta^2}} \leq \ln 0.01$$

$$\therefore -\zeta^2\pi^2 \leq \ln^2 0.01 (1 - \zeta^2)$$

$$\therefore \zeta^2(\ln^2 0.01 + \pi^2) \geq \ln^2 0.01$$

$$\therefore \zeta \geq \sqrt{\frac{\ln^2 0.01}{\ln^2 0.01 + \pi^2}}$$

$$\therefore \zeta \geq 0.83$$

time constant:

The 2% settling time $\tau_{2\%} \geq 4\tau$

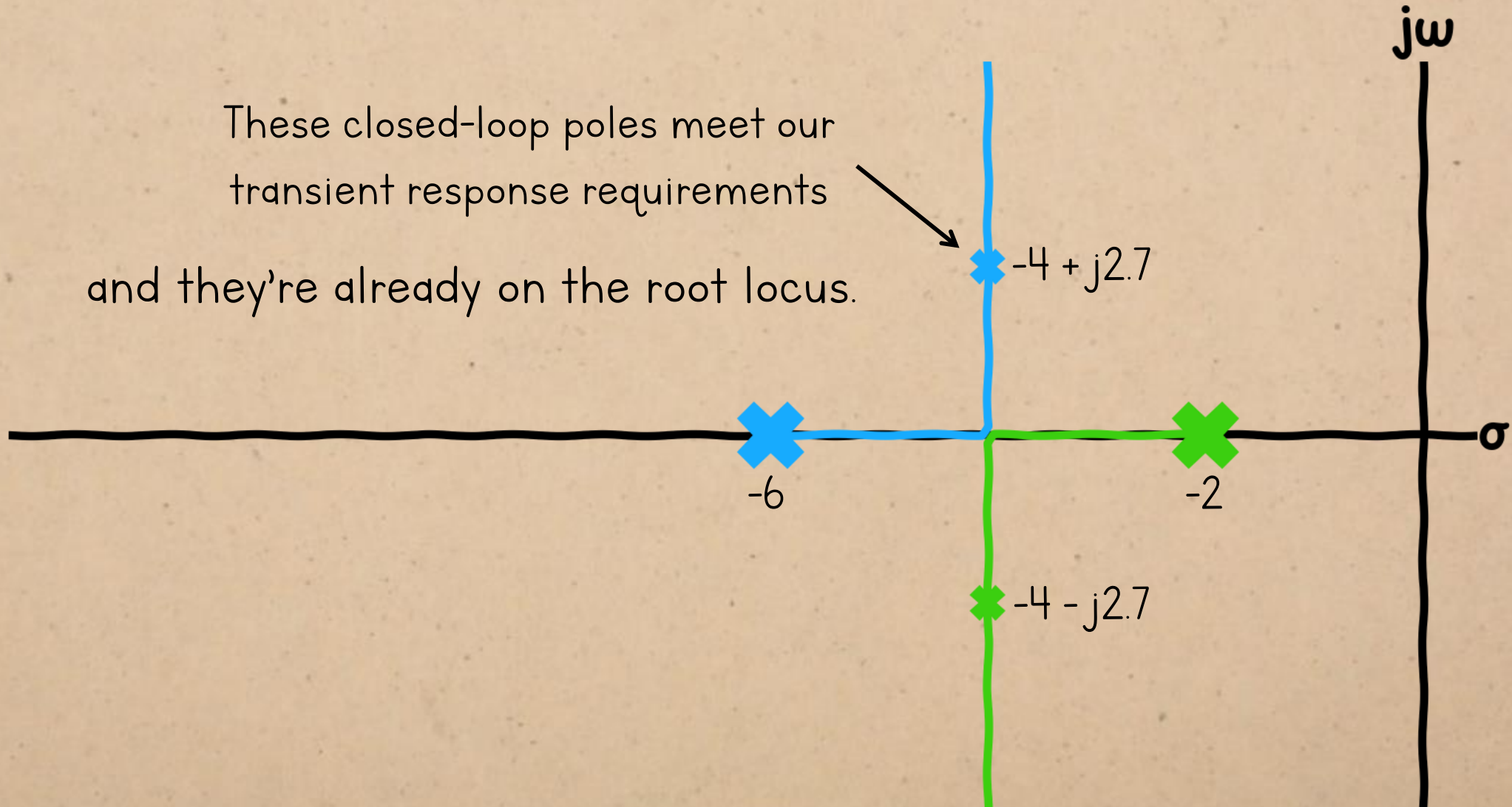
So if $\tau_{2\%} \leq 1$, $4\tau \leq 1$

$$\therefore \tau \leq \frac{1}{4}$$

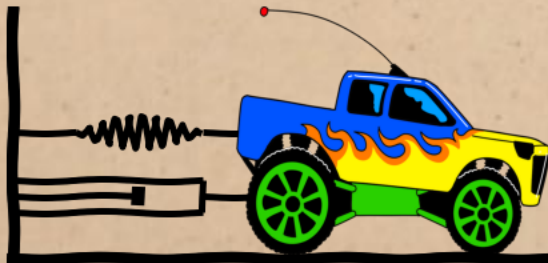
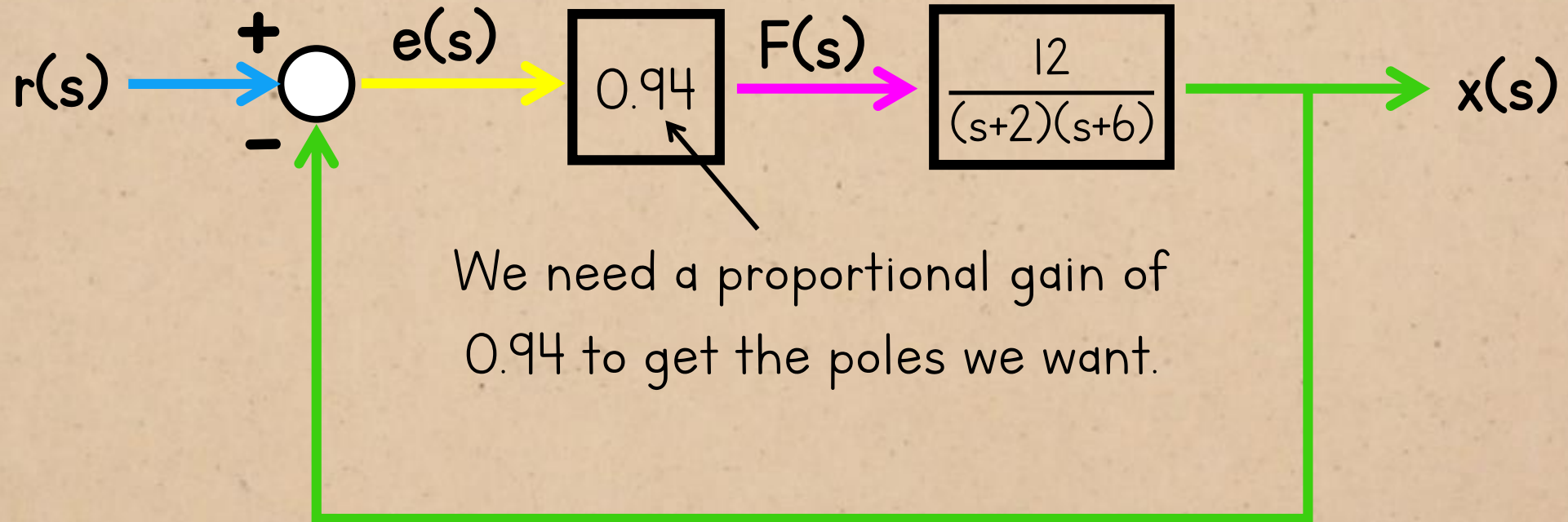
So the real part of the pole must be ≤ -4

LAG DESIGN

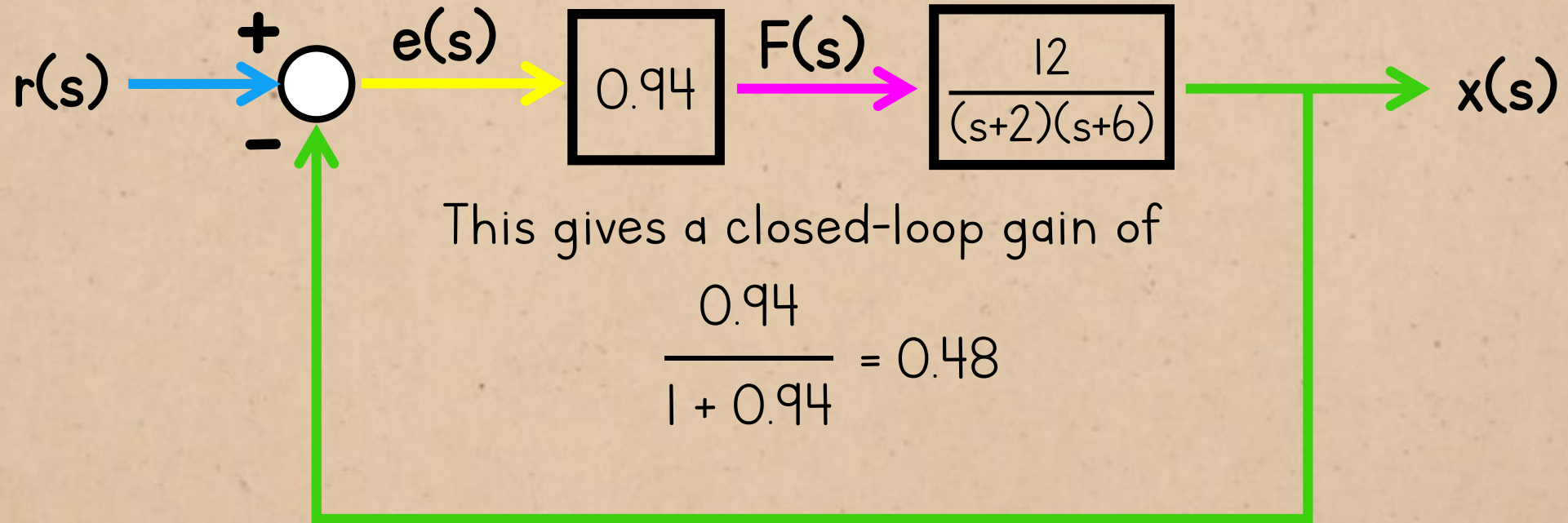
These closed-loop poles meet our transient response requirements and they're already on the root locus.



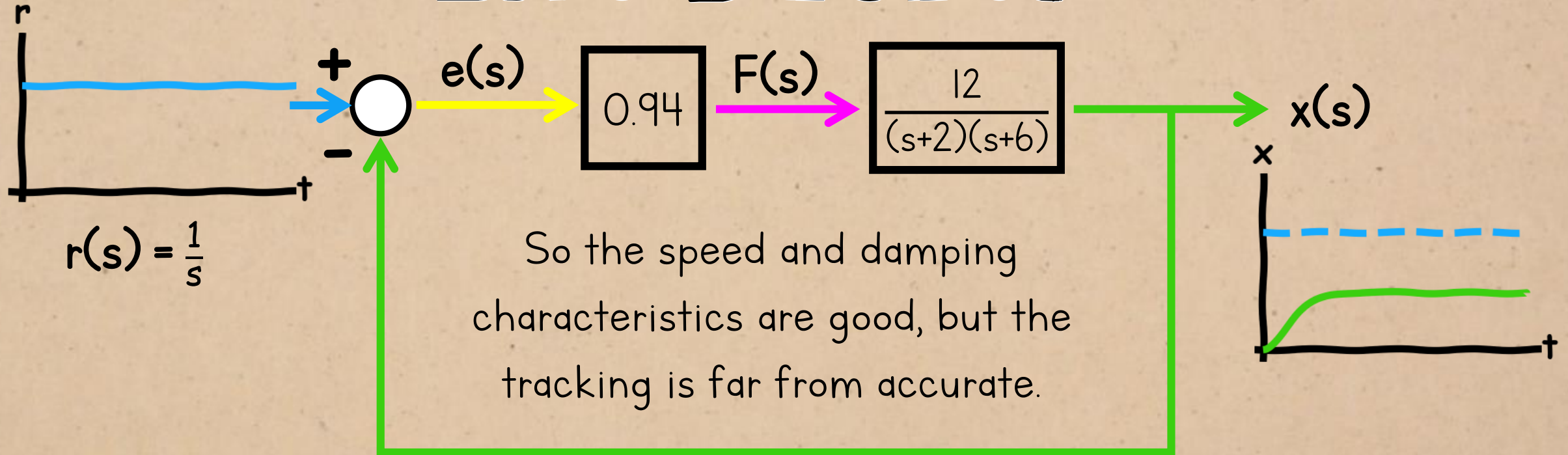
LAG DESIGN



LAG DESIGN



LAG DESIGN



LAG DESIGN

Now that we've found the proportional gain we need to get an acceptable transient response...

$$0.94 \frac{s - z}{s - p}$$

LAG DESIGN

We'll use the pole and zero to sort out the steady-state error.

$$0.94 \frac{s - z}{s - p}$$

LAG DESIGN

Their contribution to the DC gain is $\frac{z}{p}$

$$0.94 \frac{0 - z}{0 - p}$$

LAG DESIGN

Their contribution to the DC gain is $\frac{z}{p}$

$$0.94 \frac{z}{p}$$

LAG DESIGN

For the error to be $< 10\%$, we need the system's DC gain to be > 9

$$e(\infty) = \lim_{s \rightarrow 0} s \frac{1}{1 + G(s)} \frac{1}{s} \leq 0.1$$

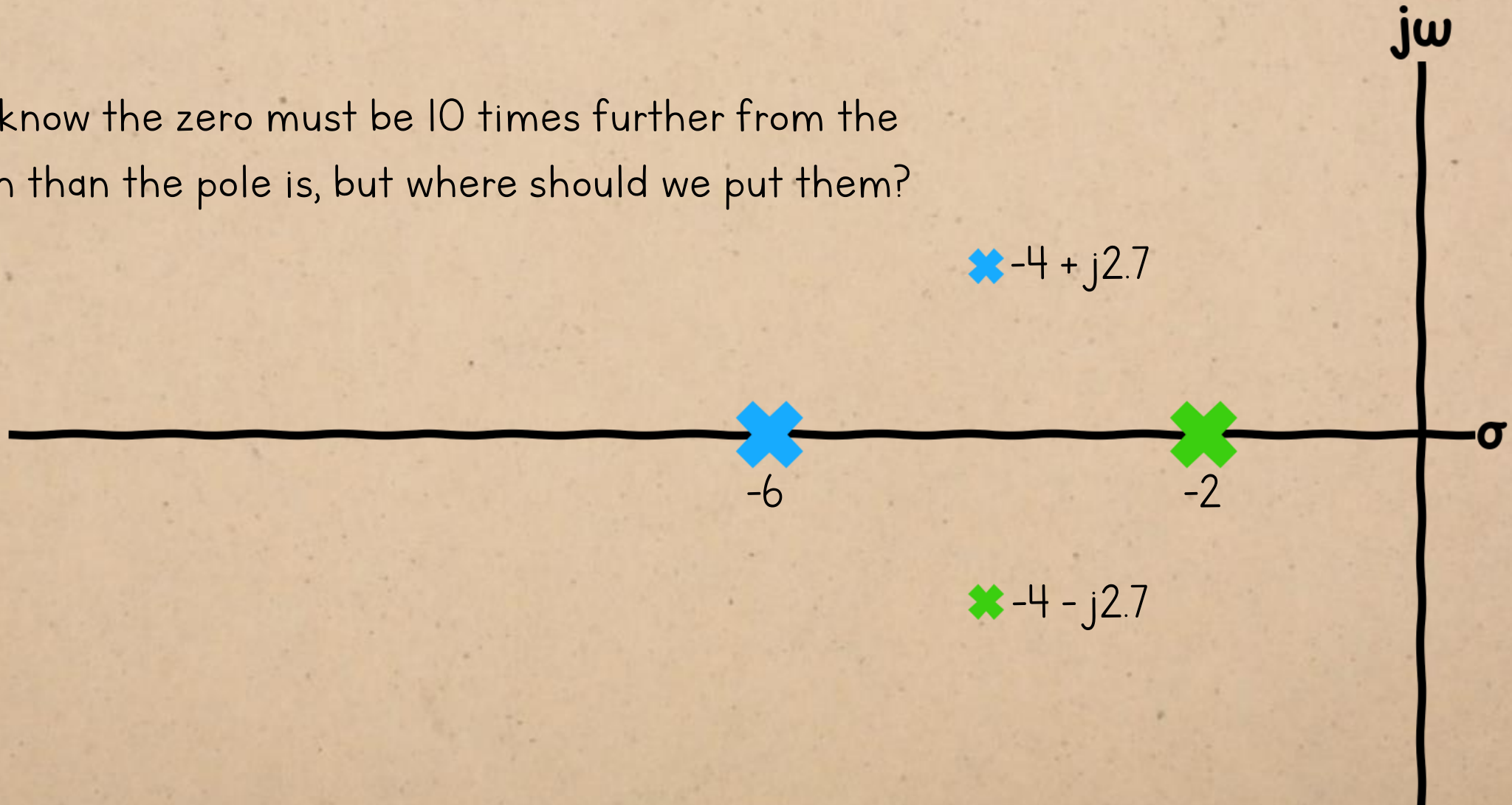
$$0.94 \frac{z}{p}$$

So $\frac{z}{p}$ must be > 9.6

(but let's say > 10)

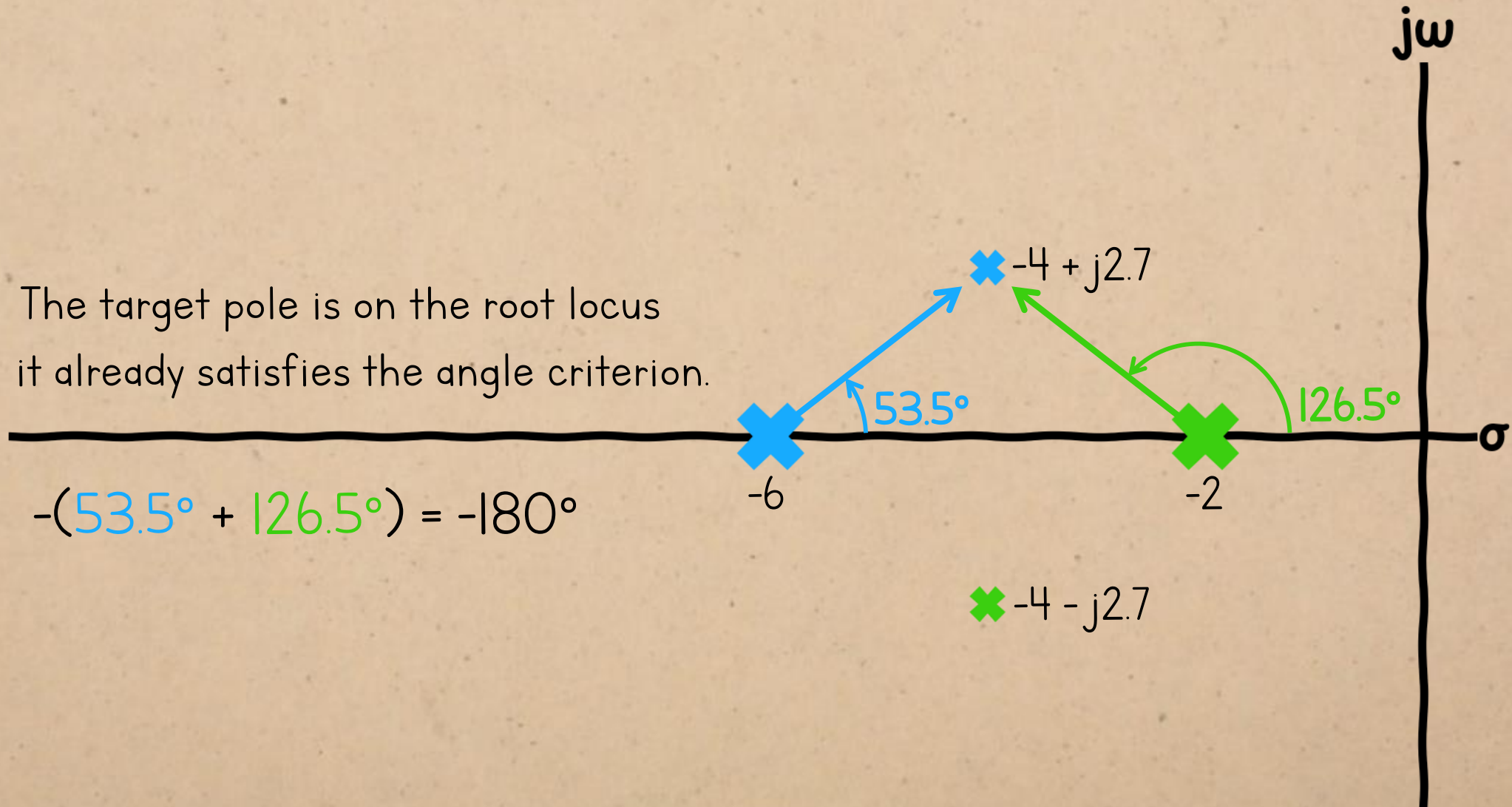
LAG DESIGN

We know the zero must be 10 times further from the origin than the pole is, but where should we put them?



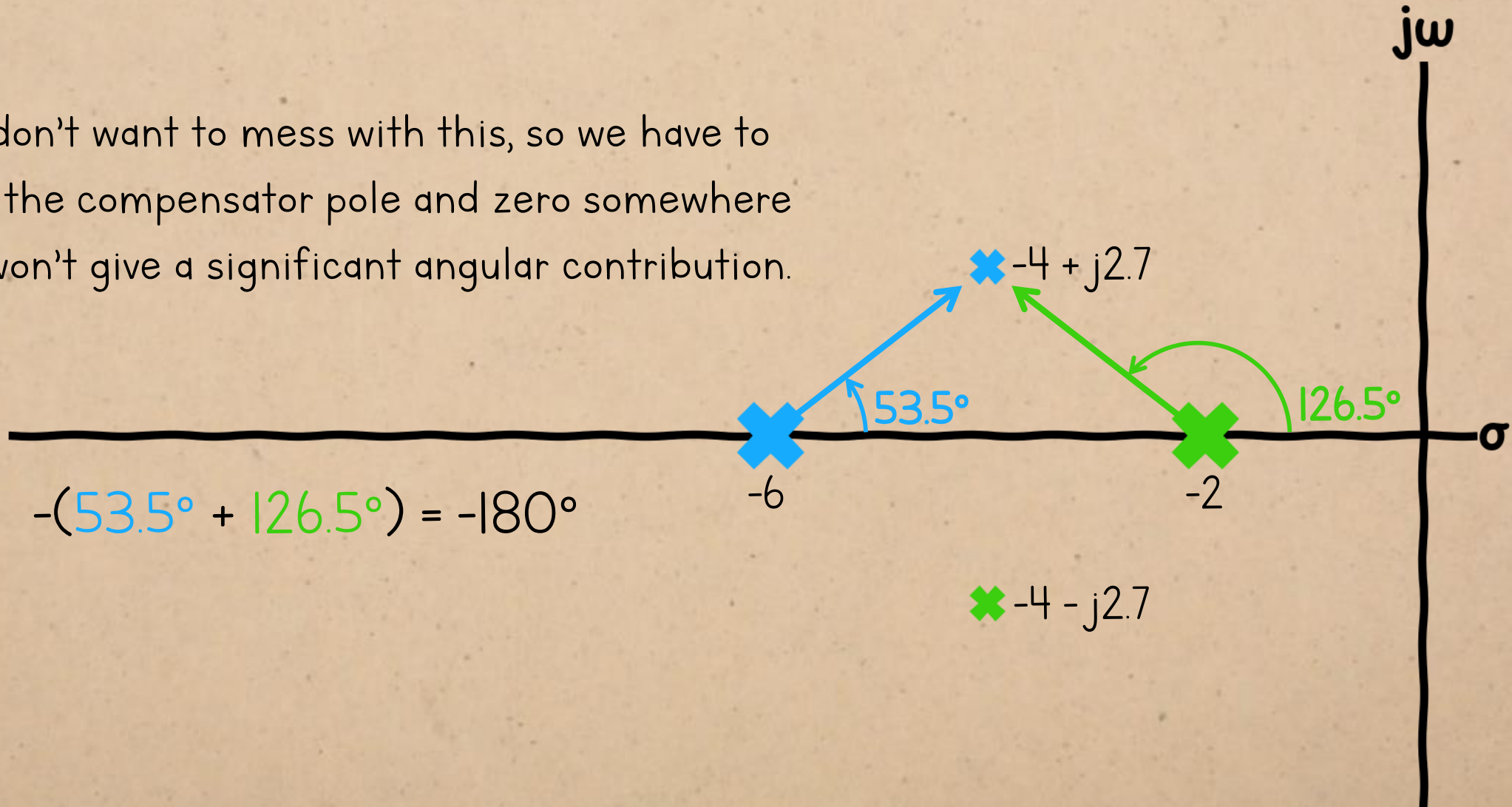
LAG DESIGN

The target pole is on the root locus
∴ it already satisfies the angle criterion.



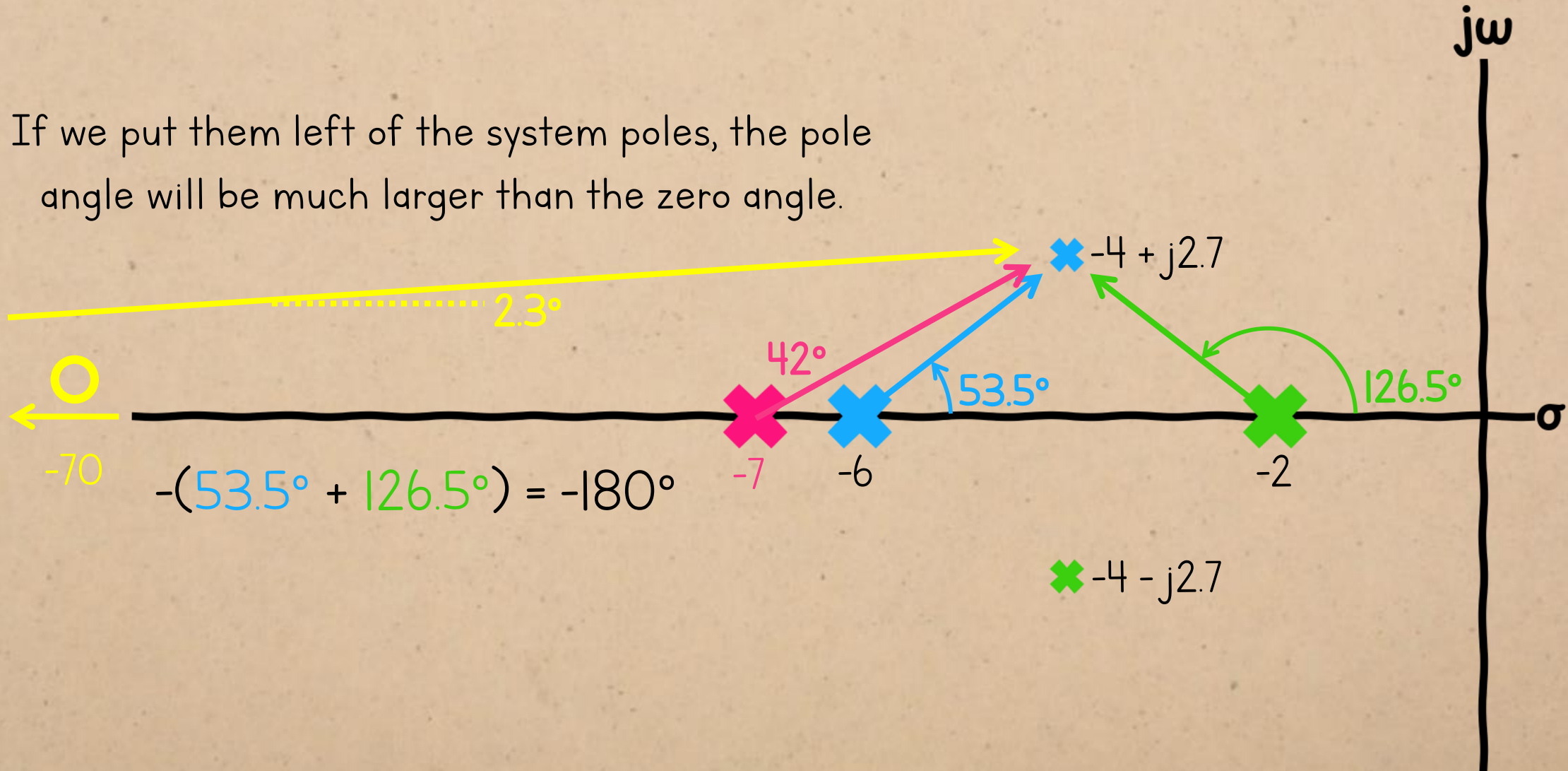
LAG DESIGN

We don't want to mess with this, so we have to place the compensator pole and zero somewhere that won't give a significant angular contribution.

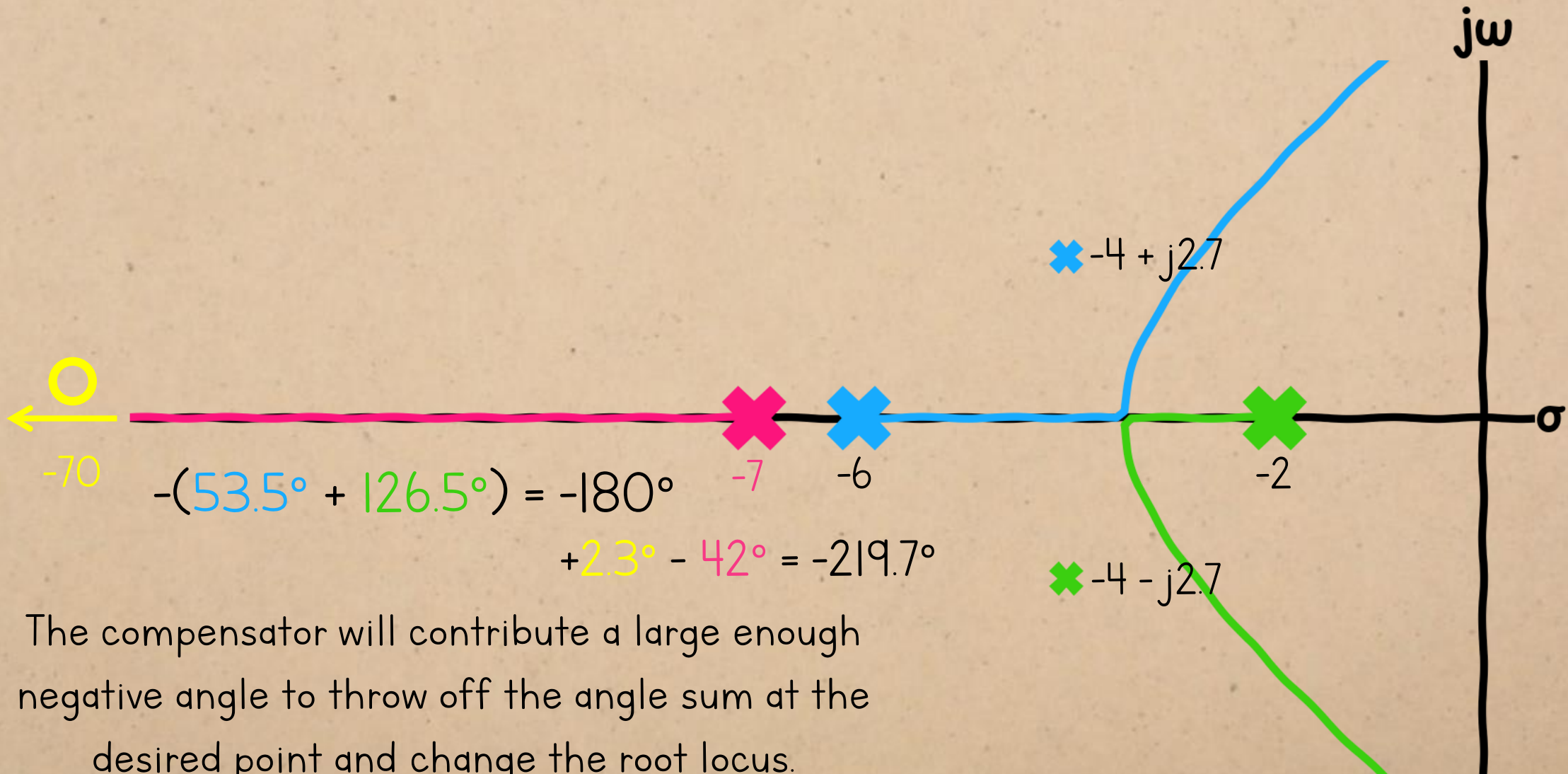


LAG DESIGN

If we put them left of the system poles, the pole angle will be much larger than the zero angle.

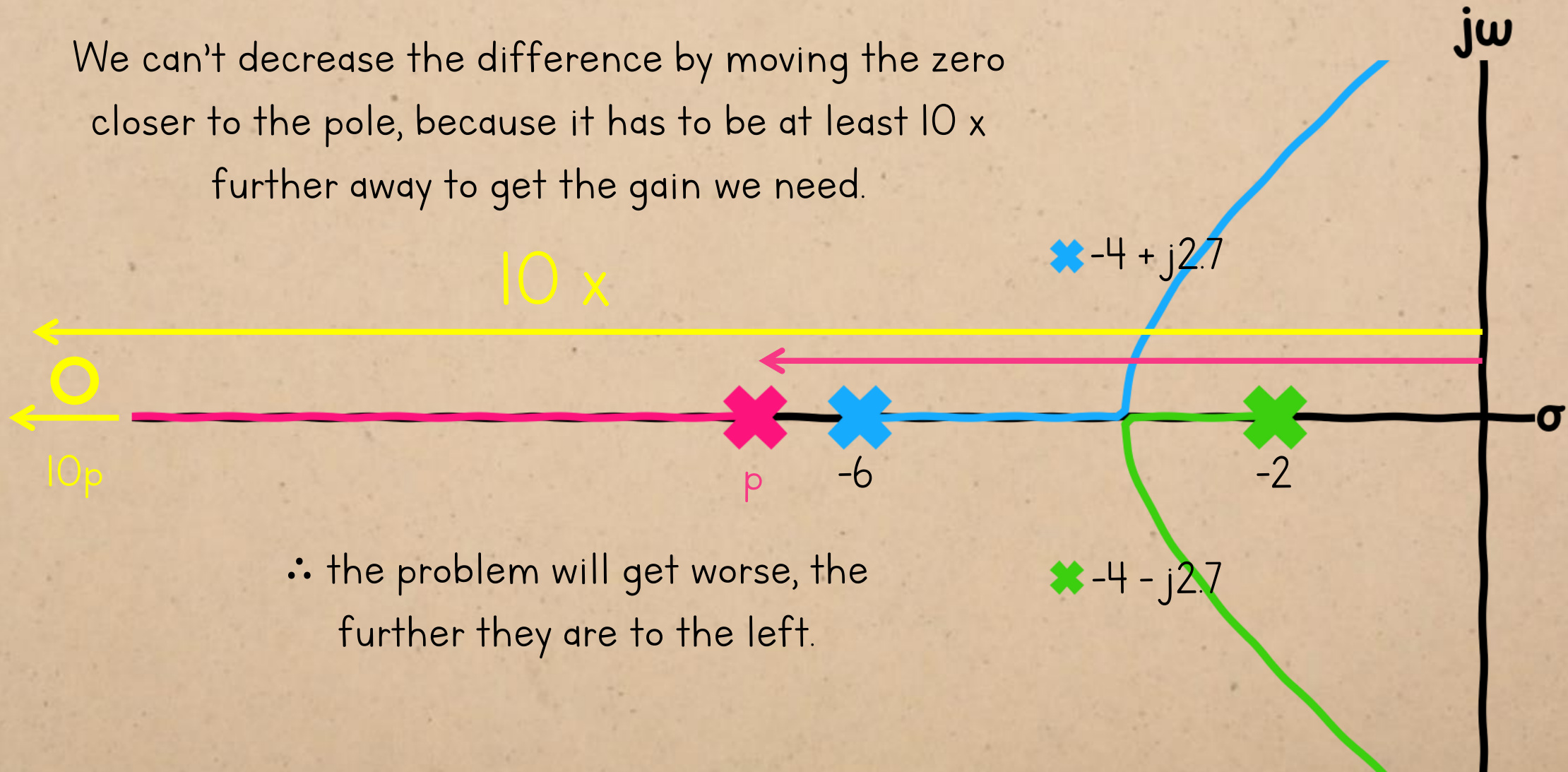


LAG DESIGN



LAG DESIGN

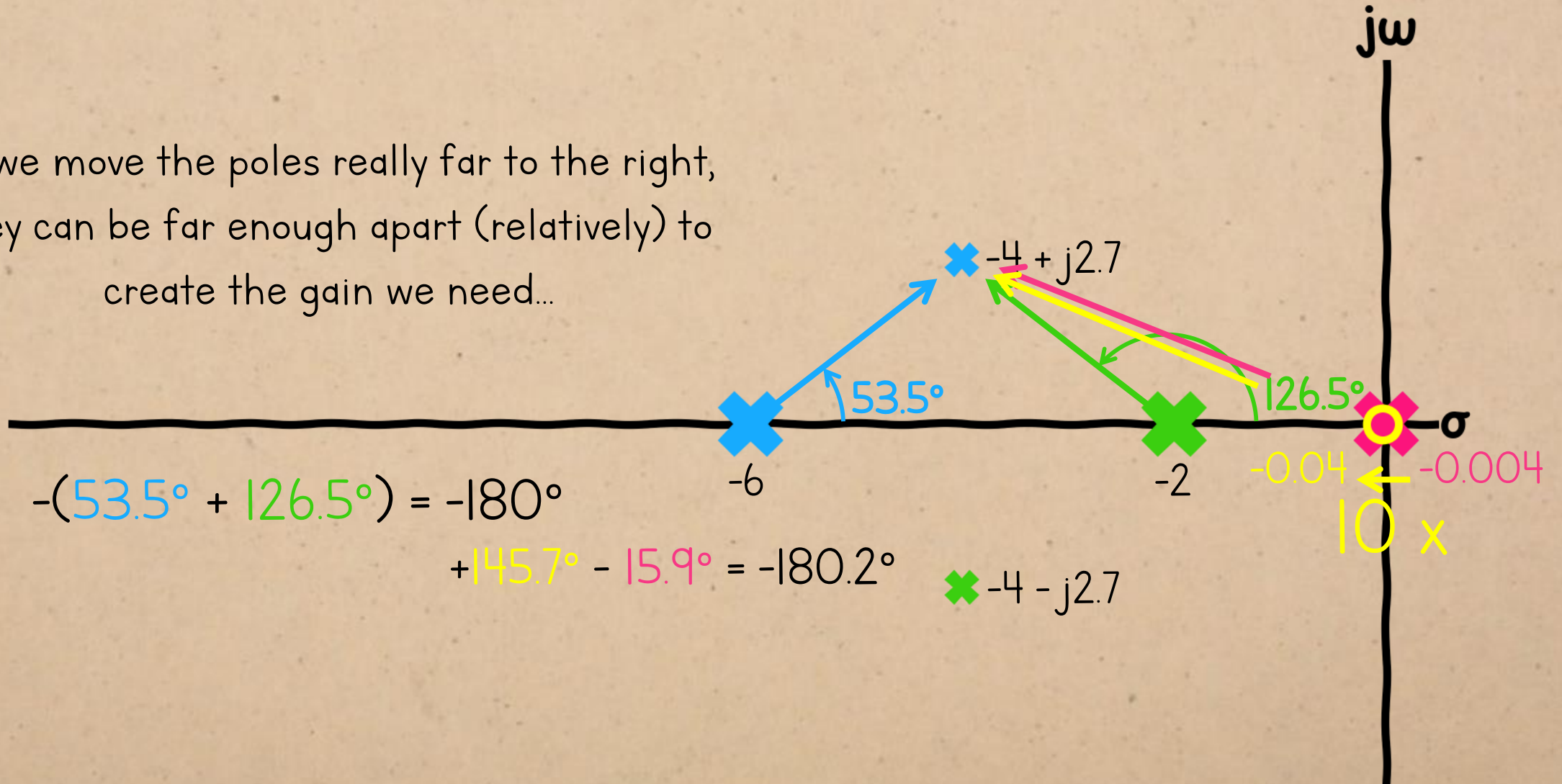
We can't decrease the difference by moving the zero closer to the pole, because it has to be at least 10 x further away to get the gain we need.



$\therefore$ the problem will get worse, the further they are to the left.

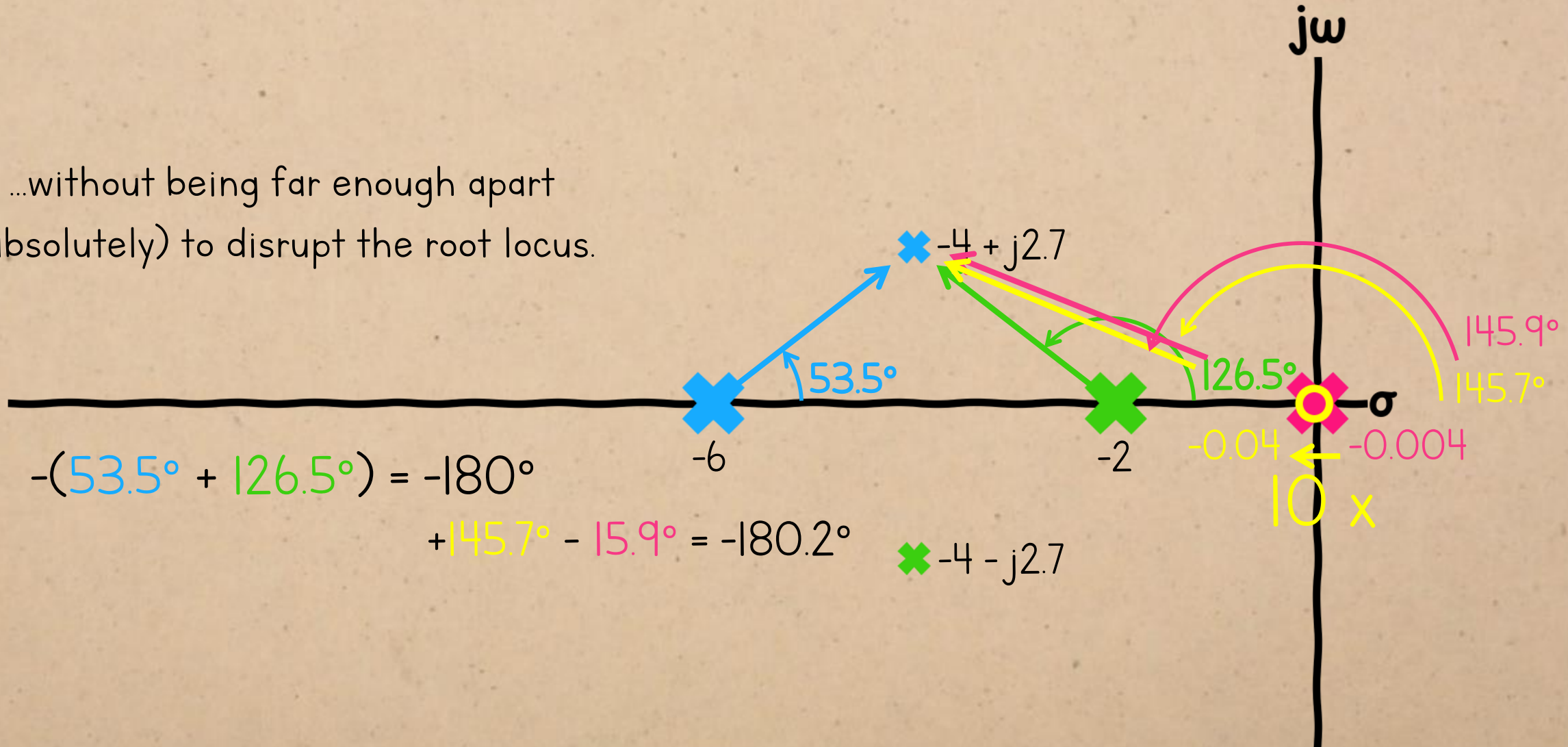
LAG DESIGN

If we move the poles really far to the right, they can be far enough apart (relatively) to create the gain we need...



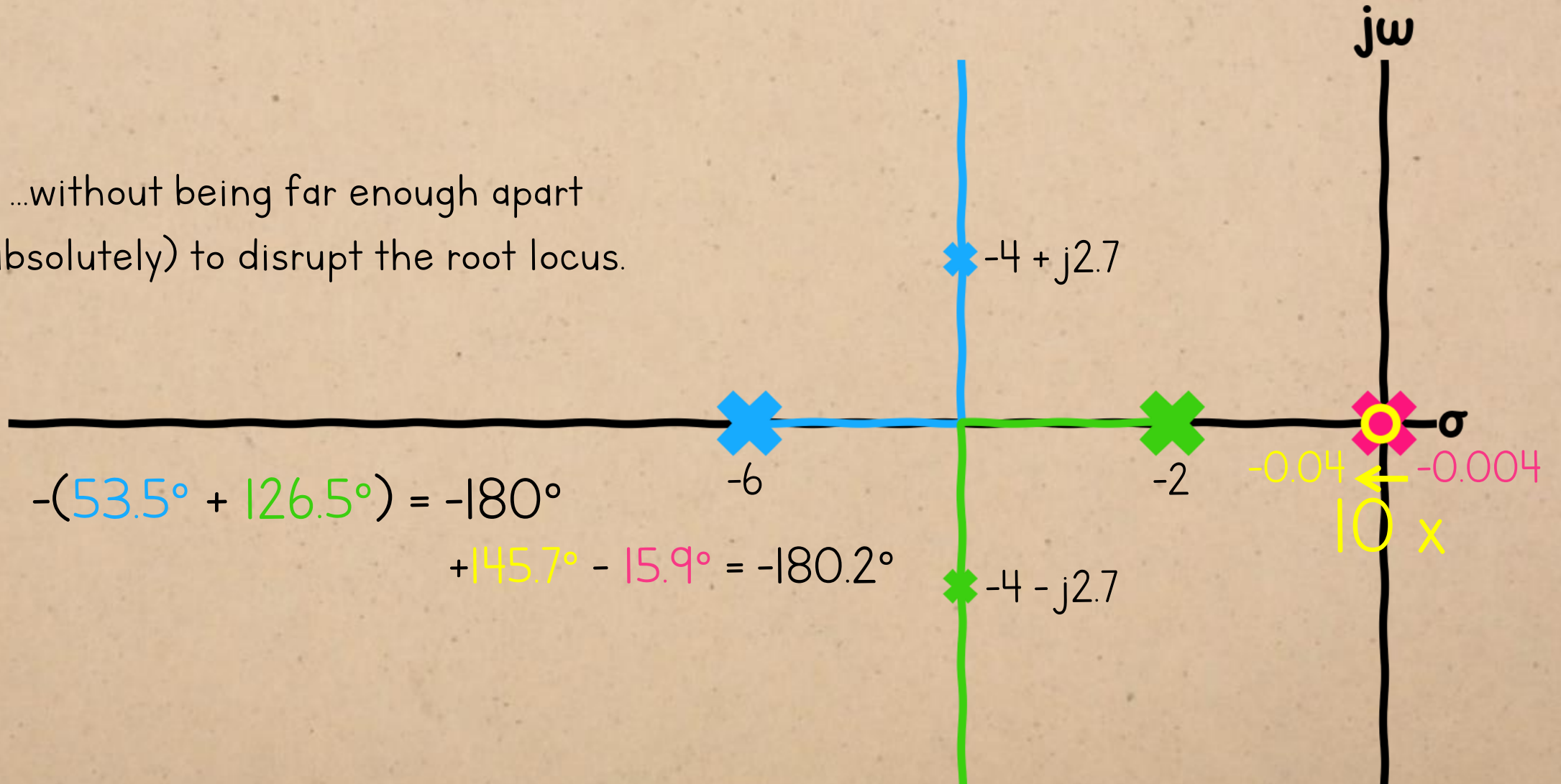
LAG DESIGN

...without being far enough apart
(absolutely) to disrupt the root locus.



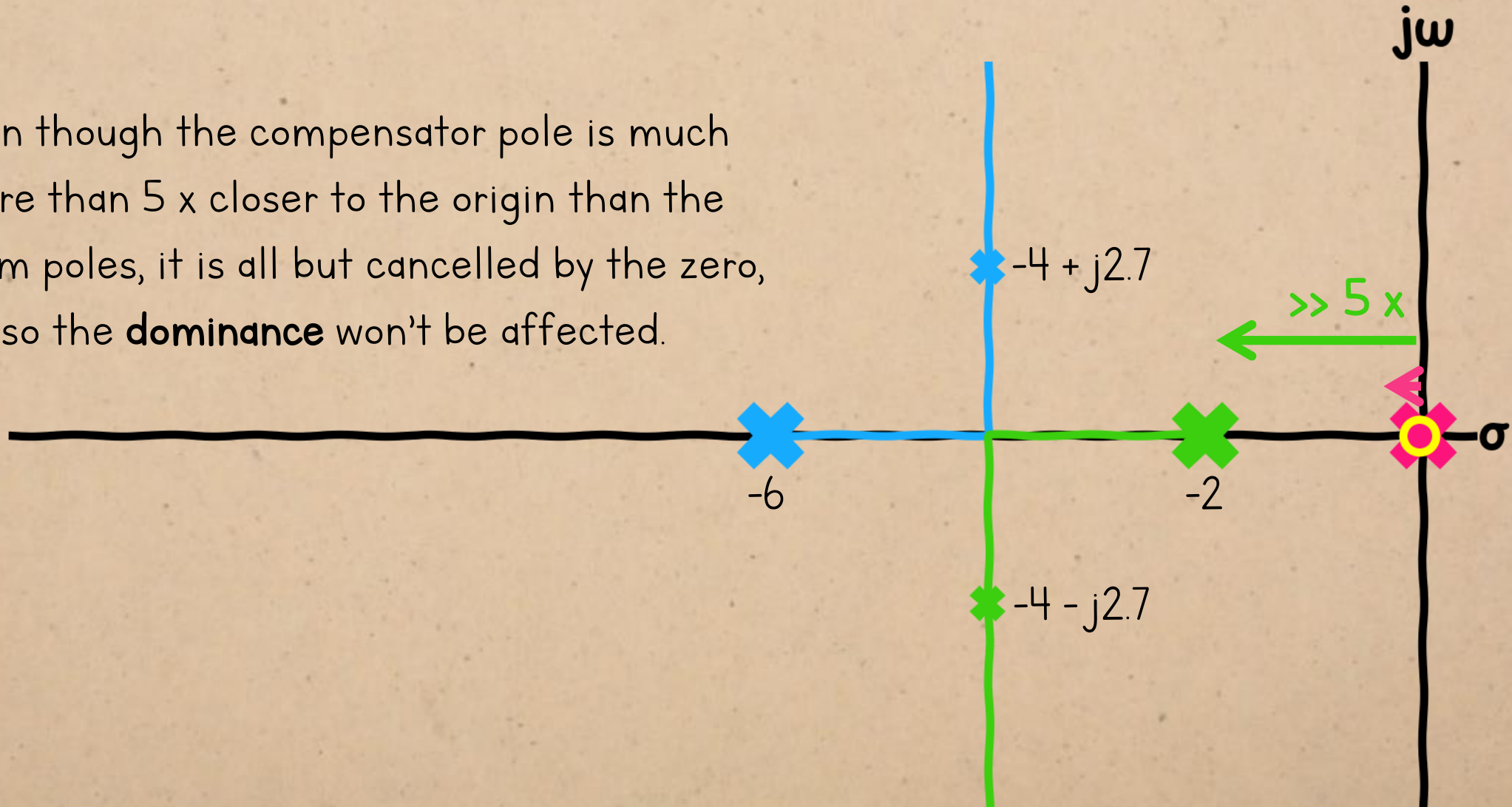
LAG DESIGN

...without being far enough apart
(absolutely) to disrupt the root locus.



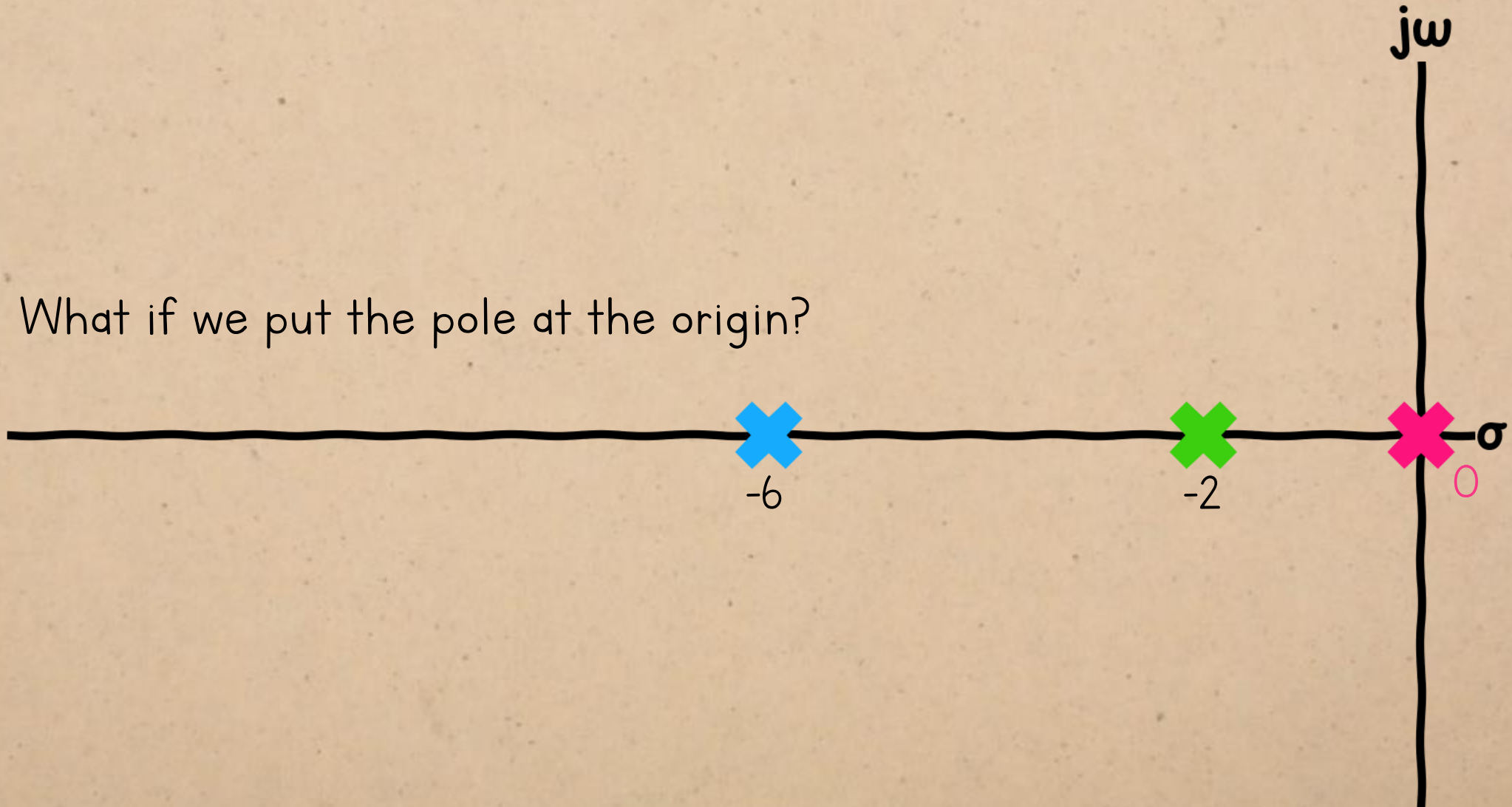
LAG DESIGN

Even though the compensator pole is much more than 5 x closer to the origin than the system poles, it is all but cancelled by the zero, so the **dominance** won't be affected.



LAG DESIGN

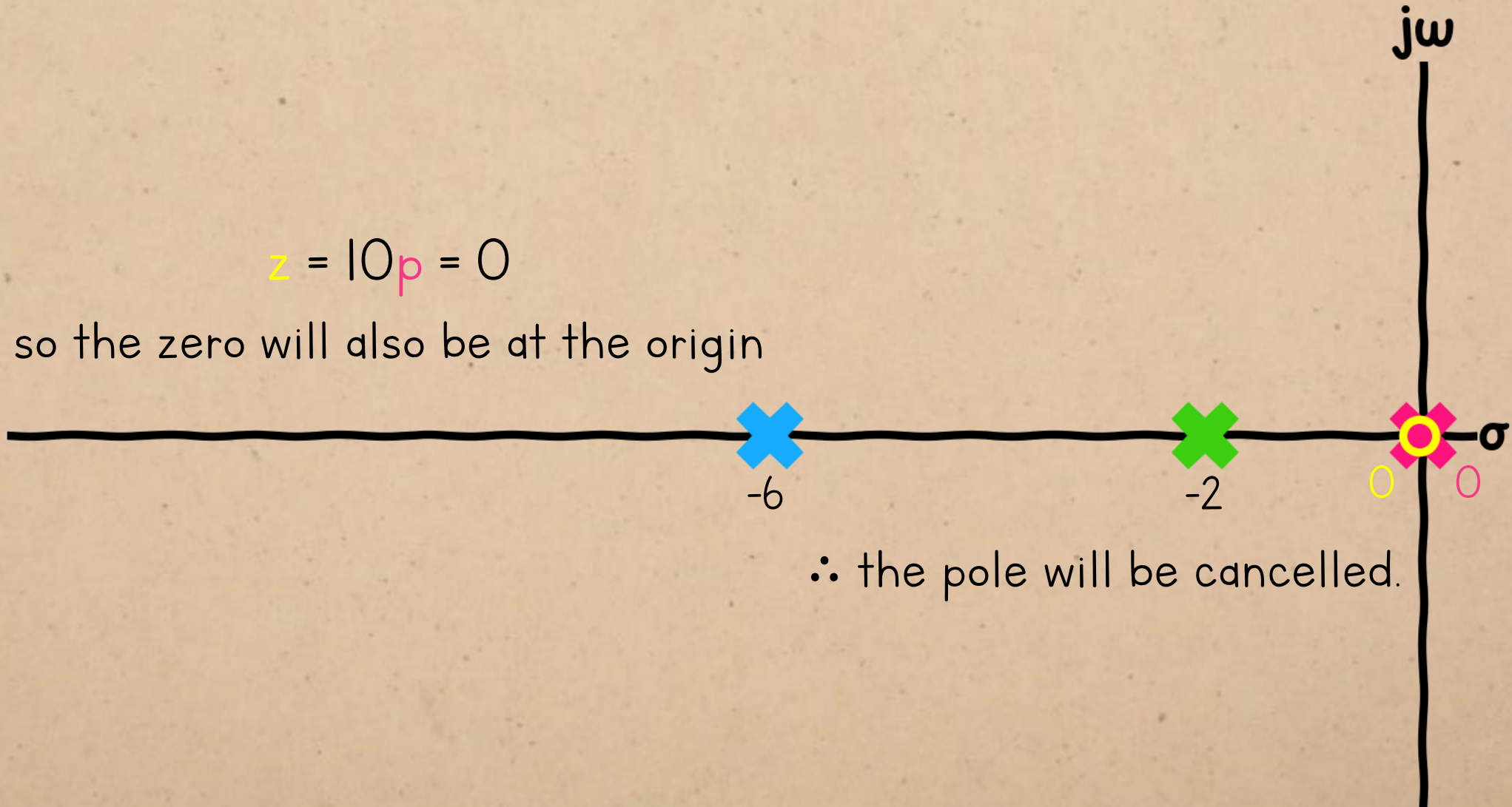
What if we put the pole at the origin?



LAG DESIGN

$$z = 10p = 0$$

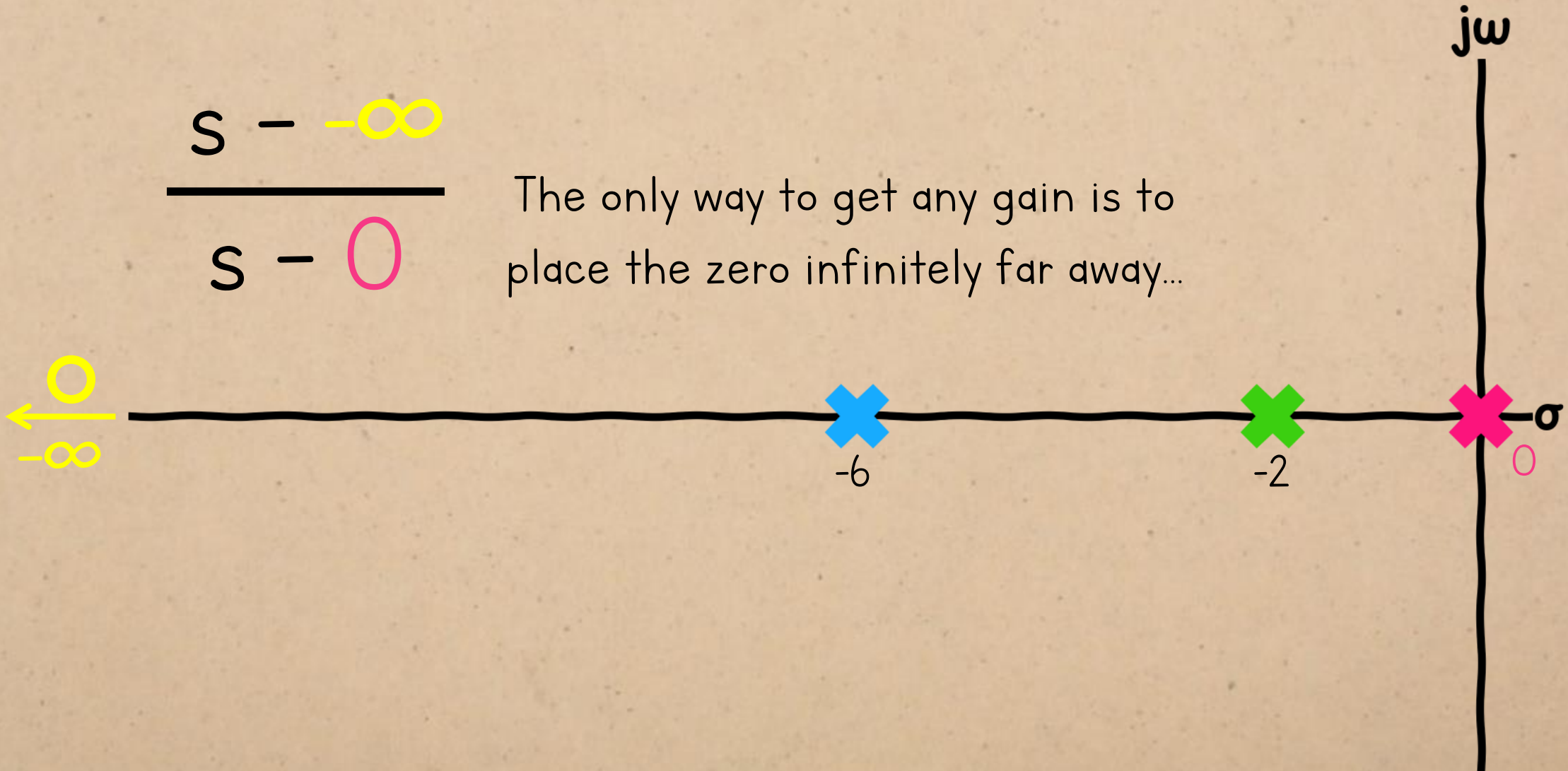
so the zero will also be at the origin



LAG DESIGN

$$\frac{s - \infty}{s - 0}$$

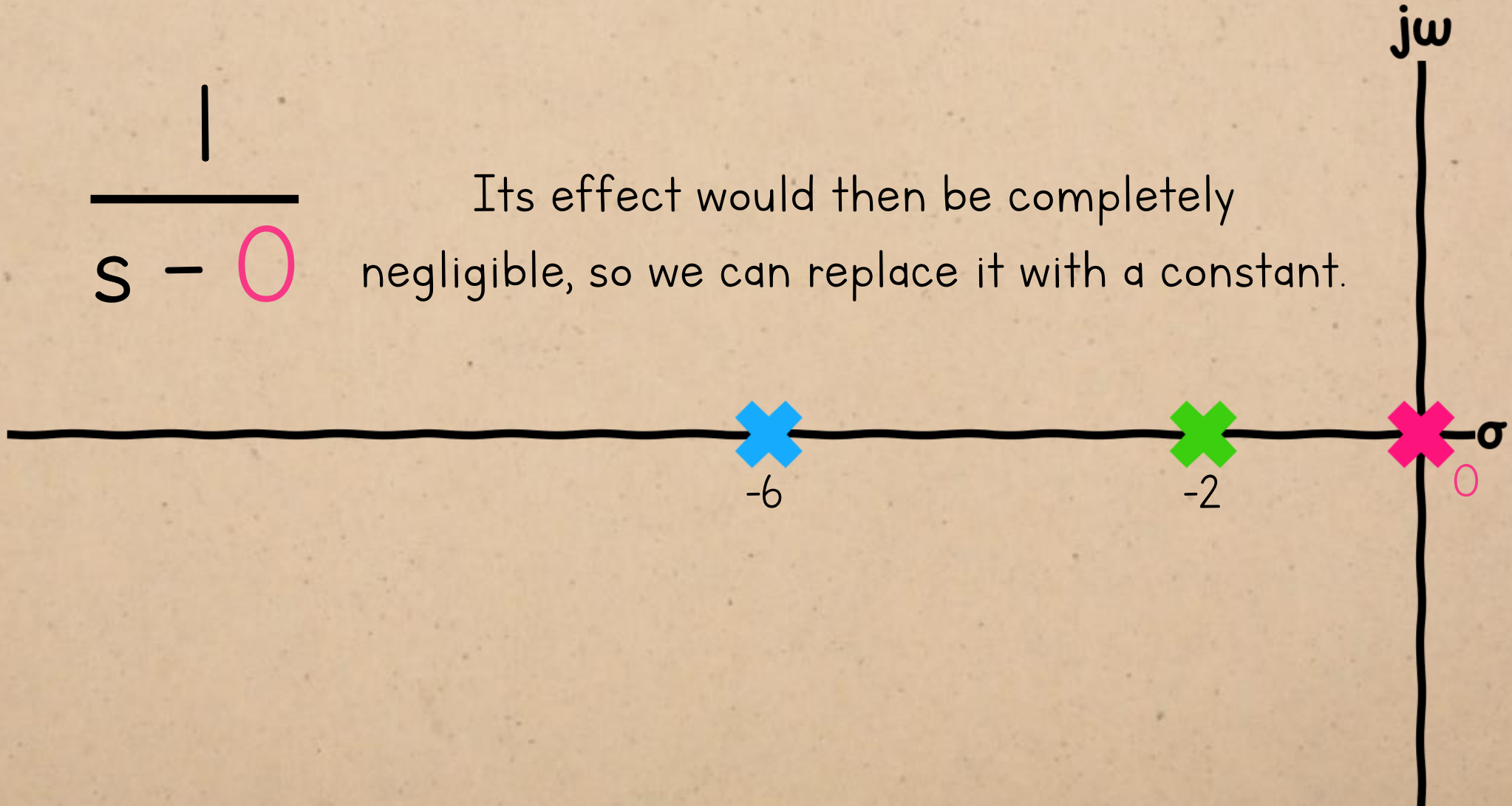
The only way to get any gain is to place the zero infinitely far away...



LAG DESIGN

$$\frac{1}{s - 0}$$

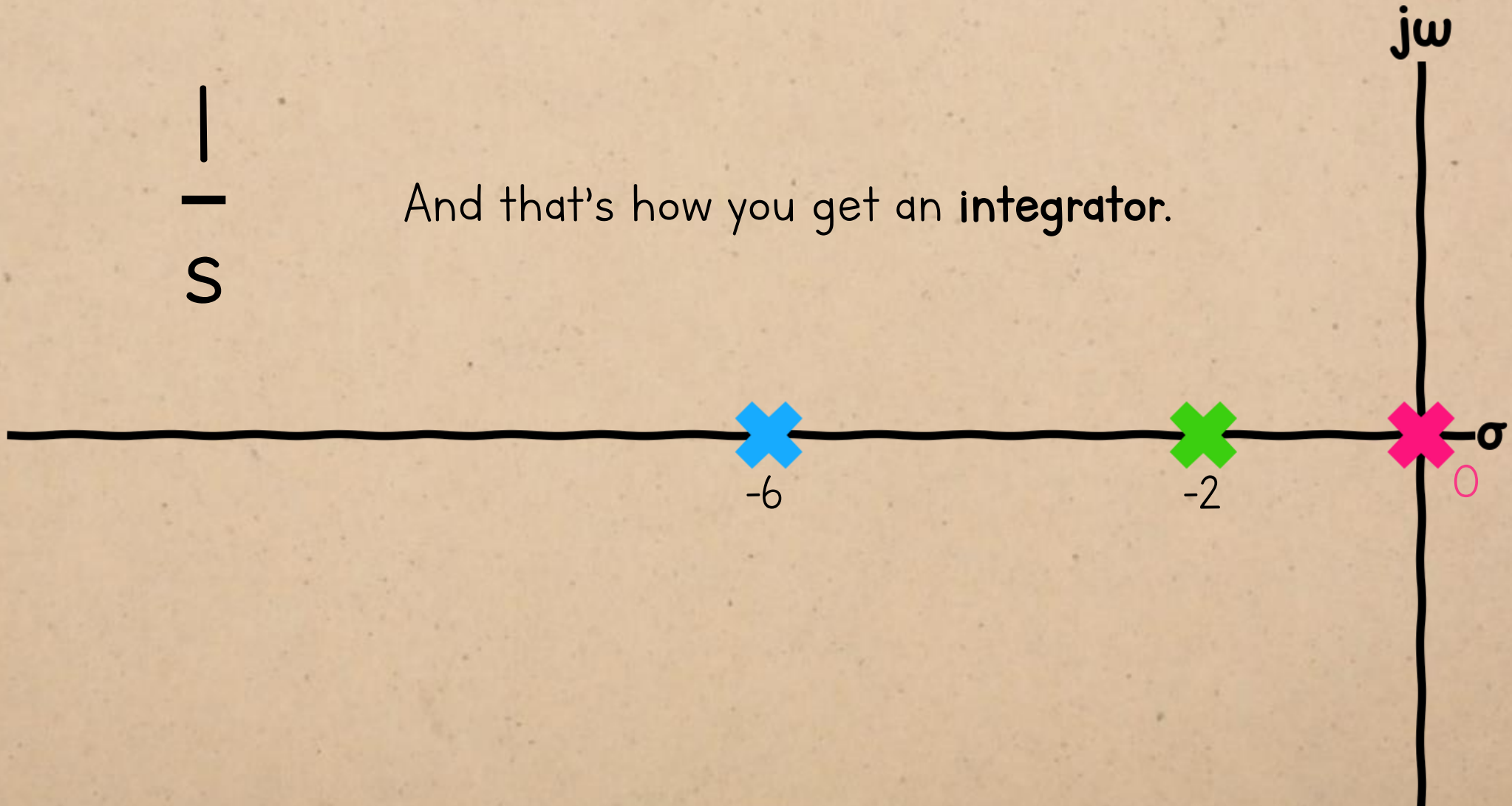
Its effect would then be completely negligible, so we can replace it with a constant.



LAG DESIGN

$$\frac{1}{s}$$

And that's how you get an **integrator**.



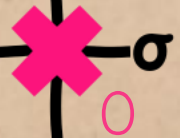


LAG DESIGN

And that's how you get an **integrator**.

Why don't we use an integrator, then? Since that would also get the steady-state error to zero.

$j\omega$



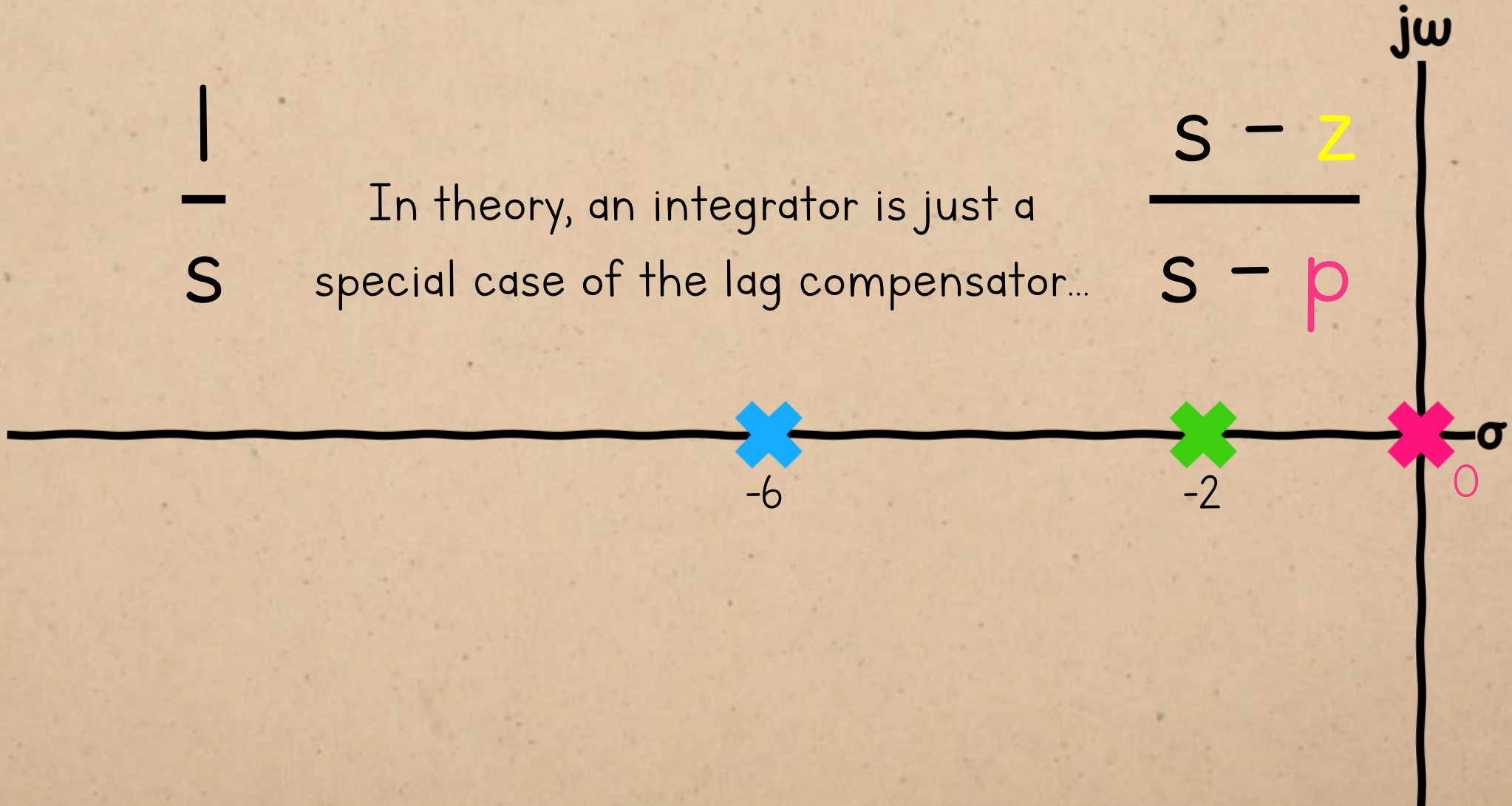
σ

LAG DESIGN

$$\frac{1}{s}$$

In theory, an integrator is just a special case of the lag compensator...

$$\frac{s - z}{s - p}$$

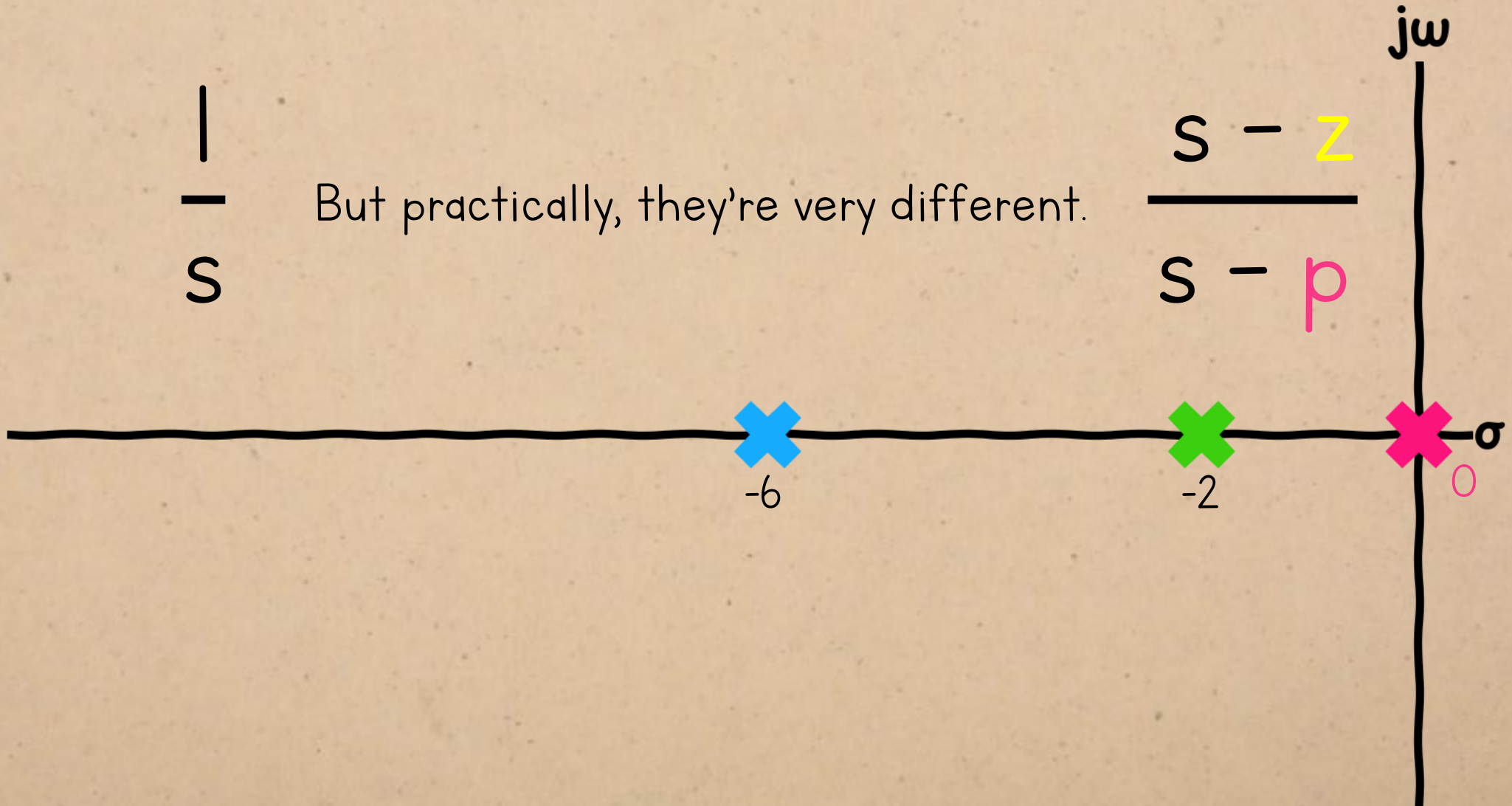


LAG DESIGN

$$\frac{1}{s}$$

But practically, they're very different.

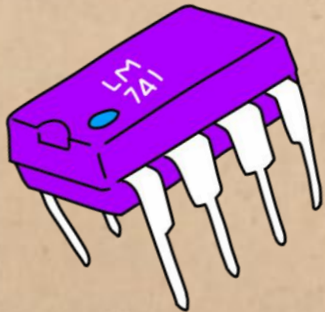
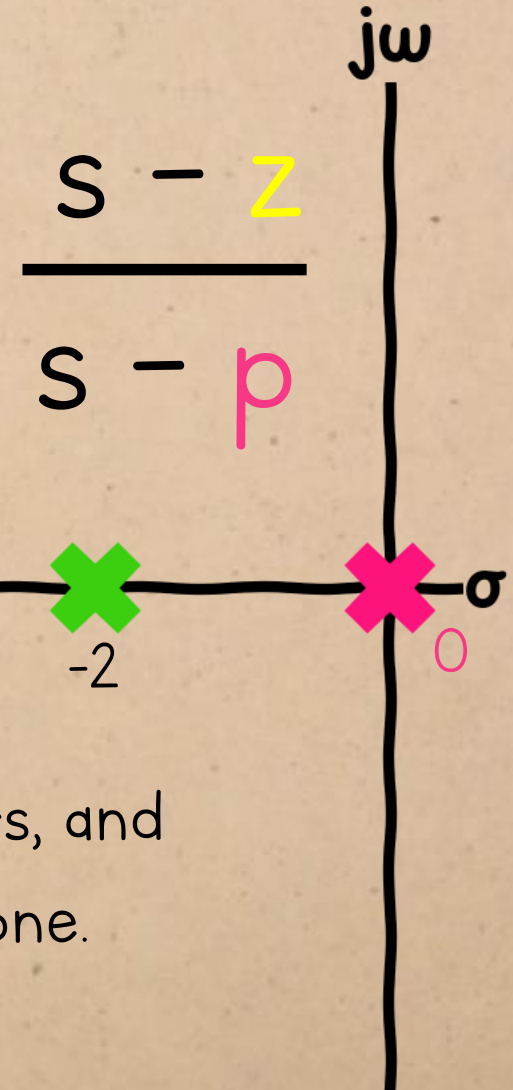
$$\frac{s - z}{s - p}$$



LAG DESIGN

$$\frac{1}{s}$$

The integrator is an **ideal** compensator i.e. has a pole at zero.

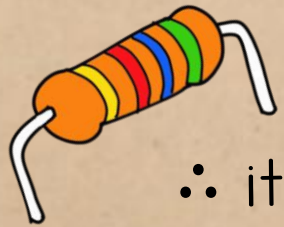
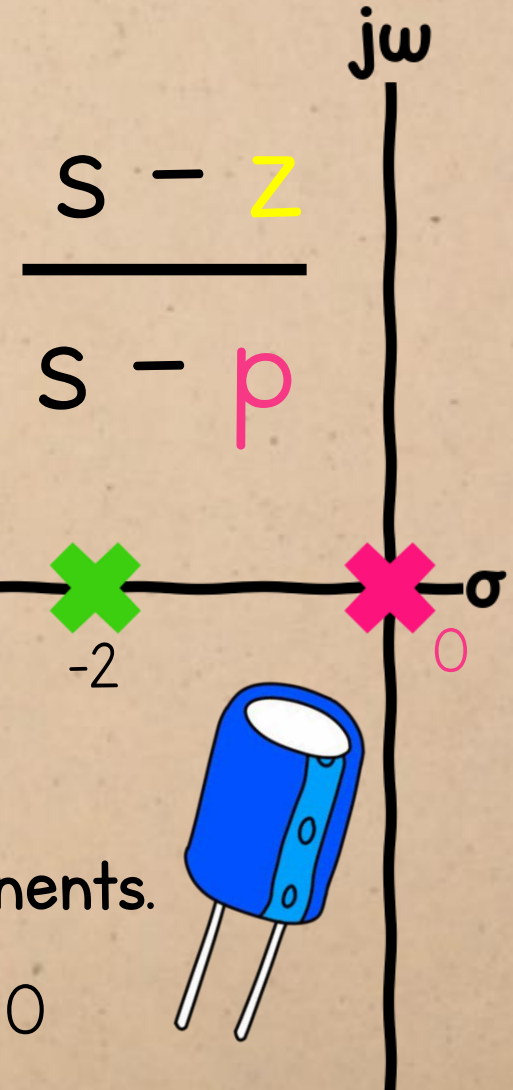


$\therefore$ you need **active components** like amplifiers, and maybe additional power sources to build one.

LAG DESIGN

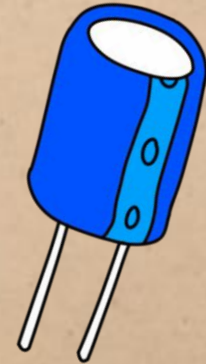
$$\frac{1}{s}$$

The lag compensator is **not ideal**
(has no poles or zeros at 0)



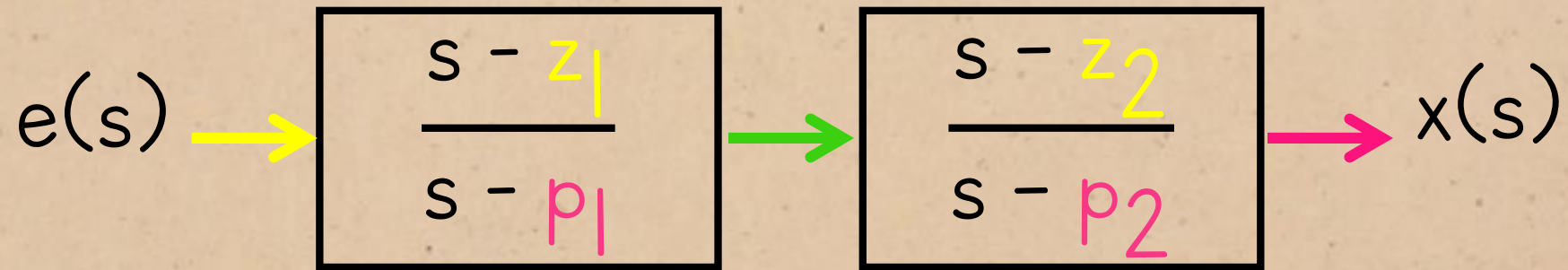
∴ it can be implemented using only **passive components**.

But these have their limitations, so we try to keep $\frac{z}{p} \leq 10$



LAG DESIGN

We can **cascade** multiple stages if we need more gain.



$$= \frac{z_1 z_2}{p_1 p_2}$$

LAG DESIGN

Where should the pole and zero go?

As close to the system
poles as possible, but
keep the angle

$< 2^\circ$

Put the zero

10 to
50 x

closer to the origin than
the system poles are.

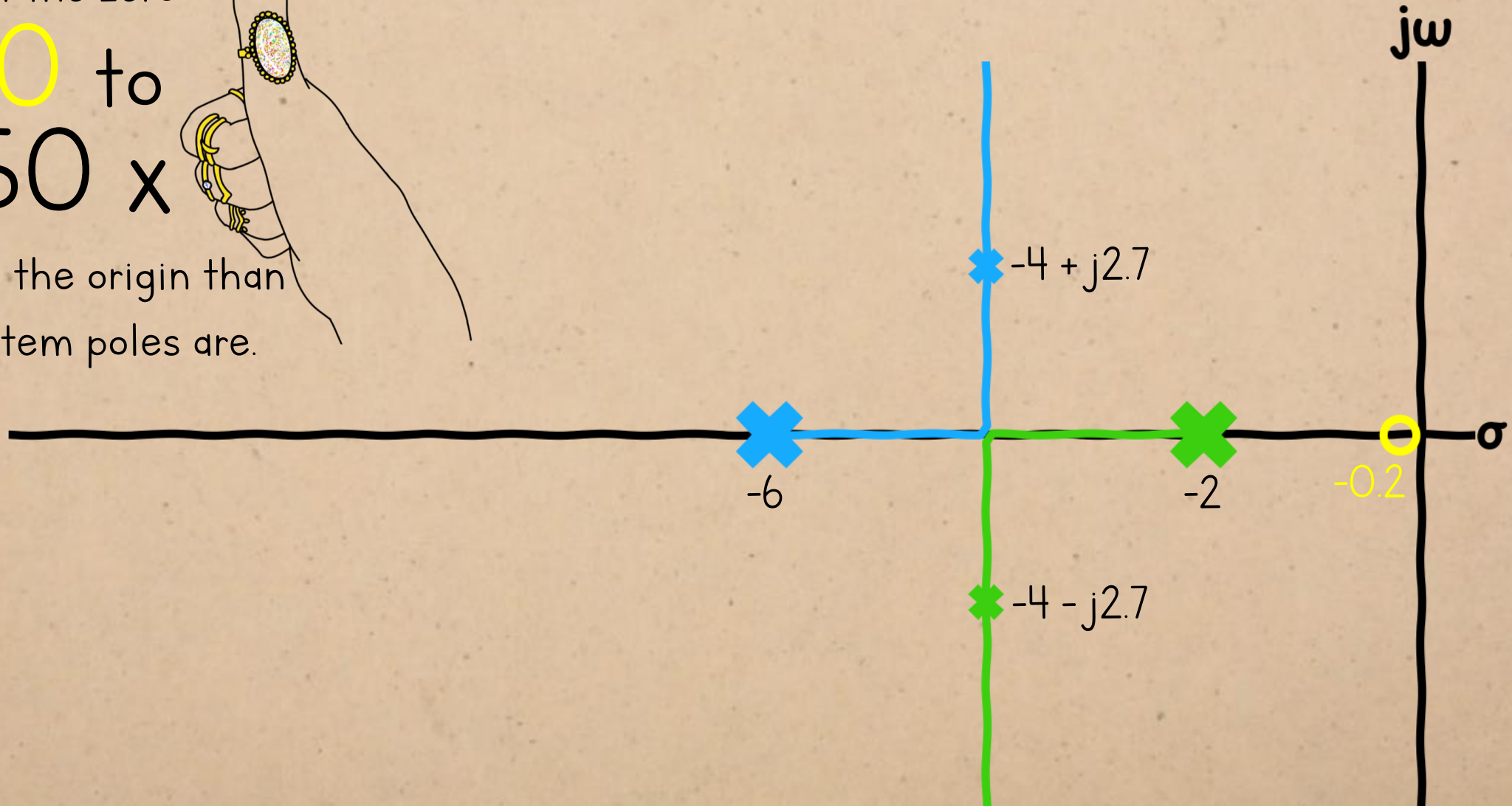
You can use both these rules of thumb together. I like to use the “10 to 50 x closer” rule to tell me where to look, and the $< 2^\circ$ rule to tell me if a point in that range that I try is acceptable.

LAG DESIGN

Put the zero

10 to
50 x

closer to the origin than
the system poles are.

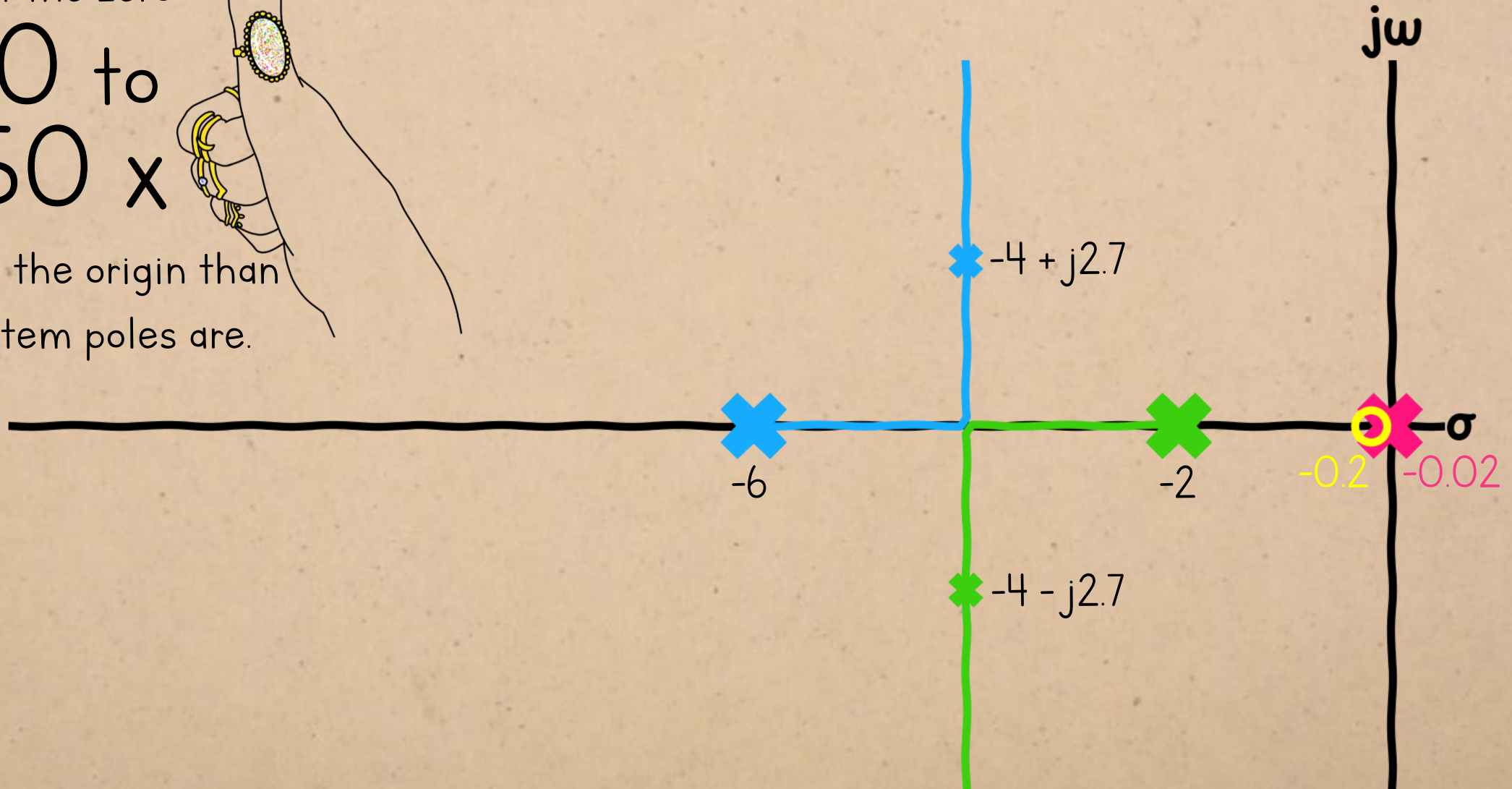
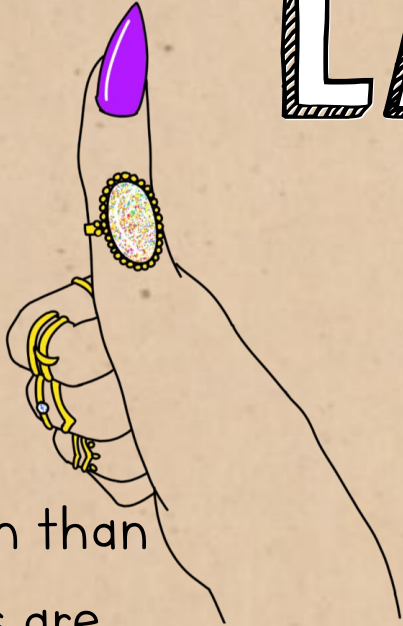


LAG DESIGN

Put the zero

10 to
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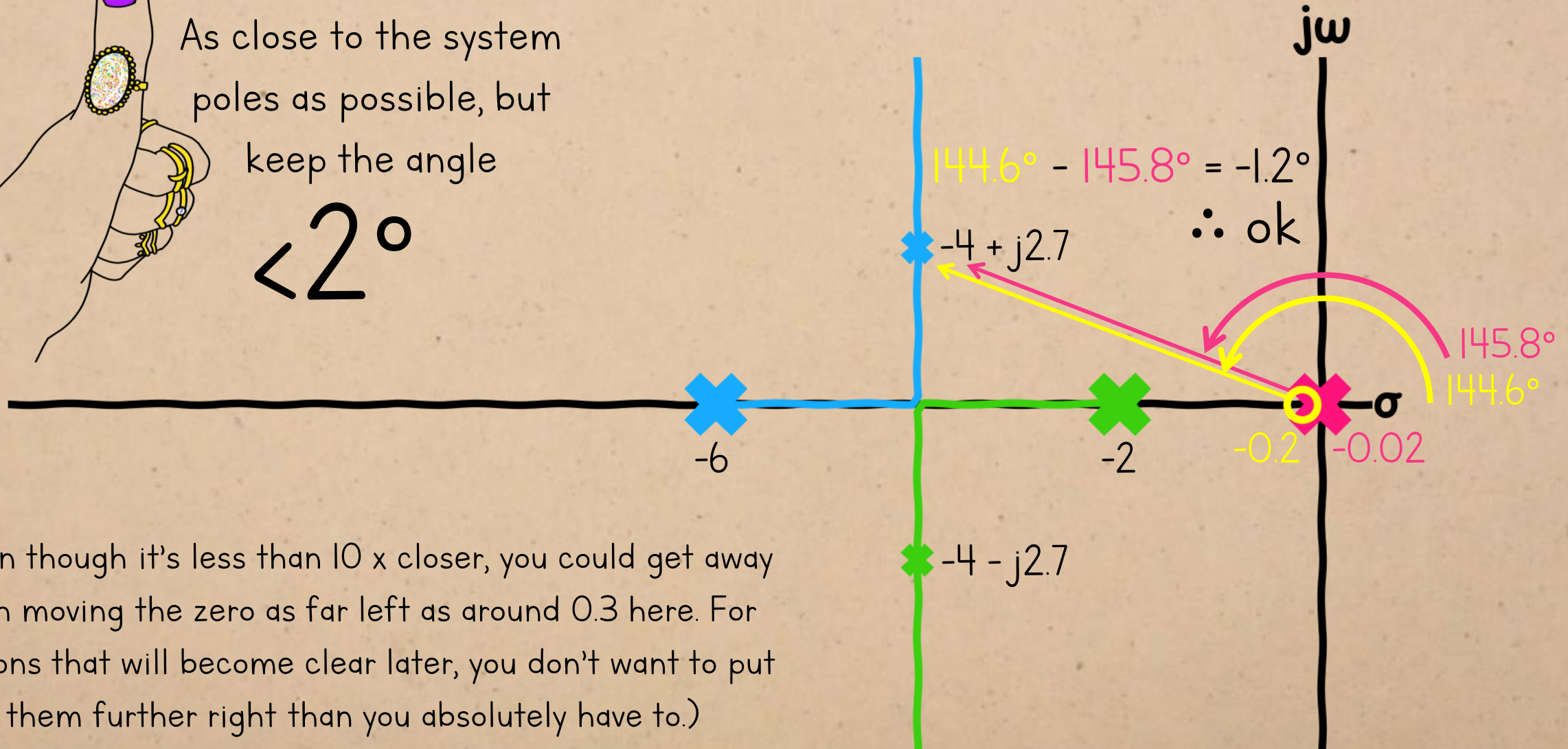


LAG DESIGN



As close to the system poles as possible, but keep the angle

$$< 2^\circ$$



(Even though it's less than 10 x closer, you could get away with moving the zero as far left as around 0.3 here. For reasons that will become clear later, you don't want to put them further right than you absolutely have to.)

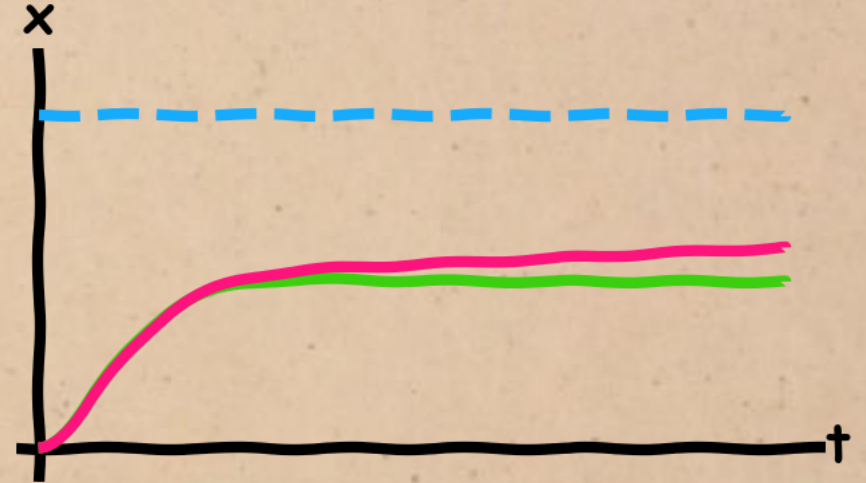
TESTING

$$0.94 \frac{s - 0.2}{s - 0.02}$$

TESTING

C'mon,
do something...

(uncompensated)



After (lag compensator)

The system quickly gets to the
point it did before compensation,
but then moves really slowly.

TESTING

Before (uncompensated)



After (lag compensator)



It does eventually get within 90% of the finish line, like it's supposed to, but it takes ages.

TESTING

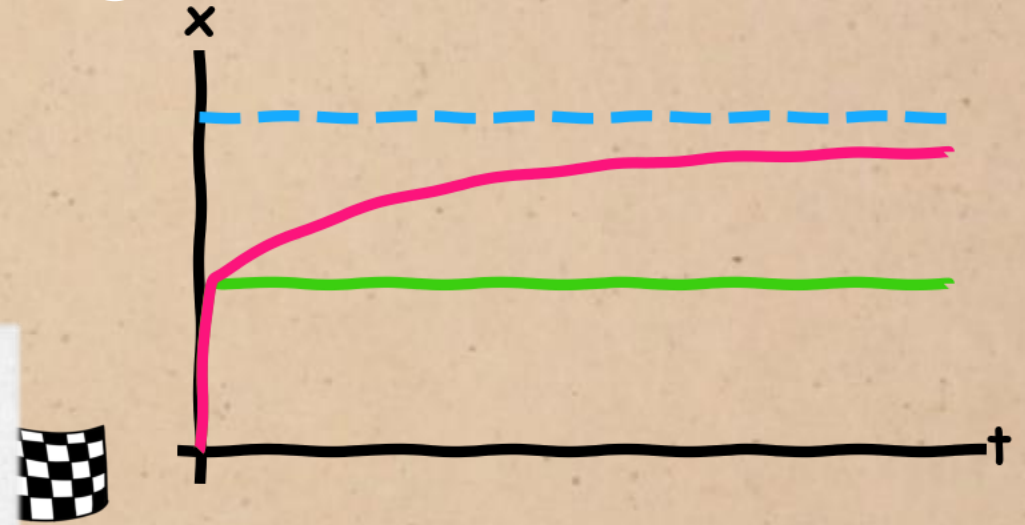
Before (uncompensated)

design a controller to meet the following requirements:

~~1. Position error $< 10\%$~~

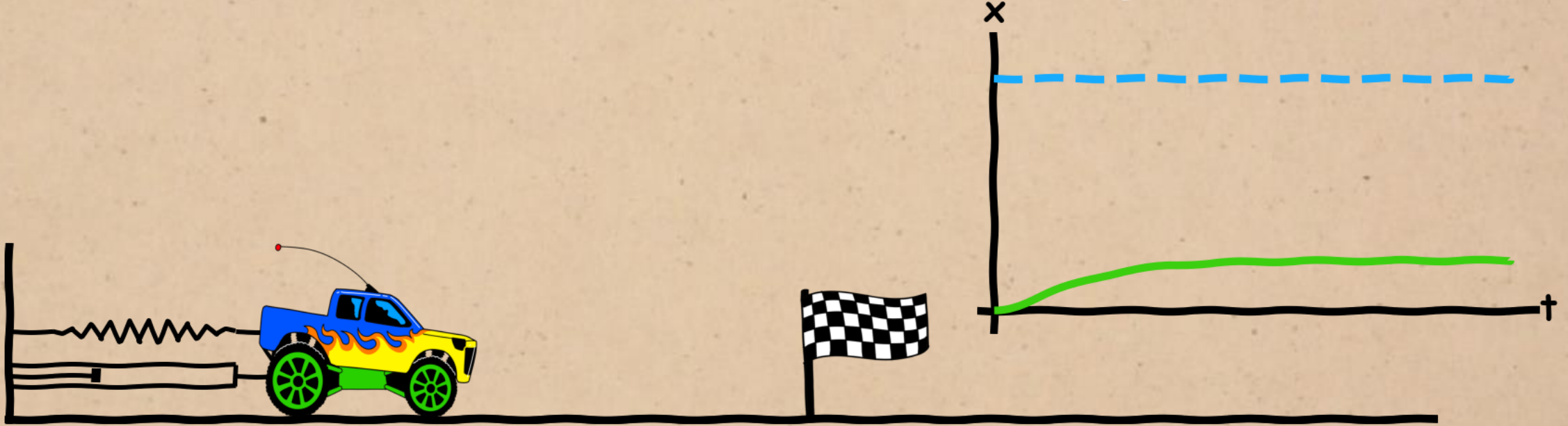
~~2. % overshoot $< 1\%$~~

3. Settling time to 2% < 1 second ?



Why didn't it meet all the requirements?

WHAT HAPPENED?



Remember...

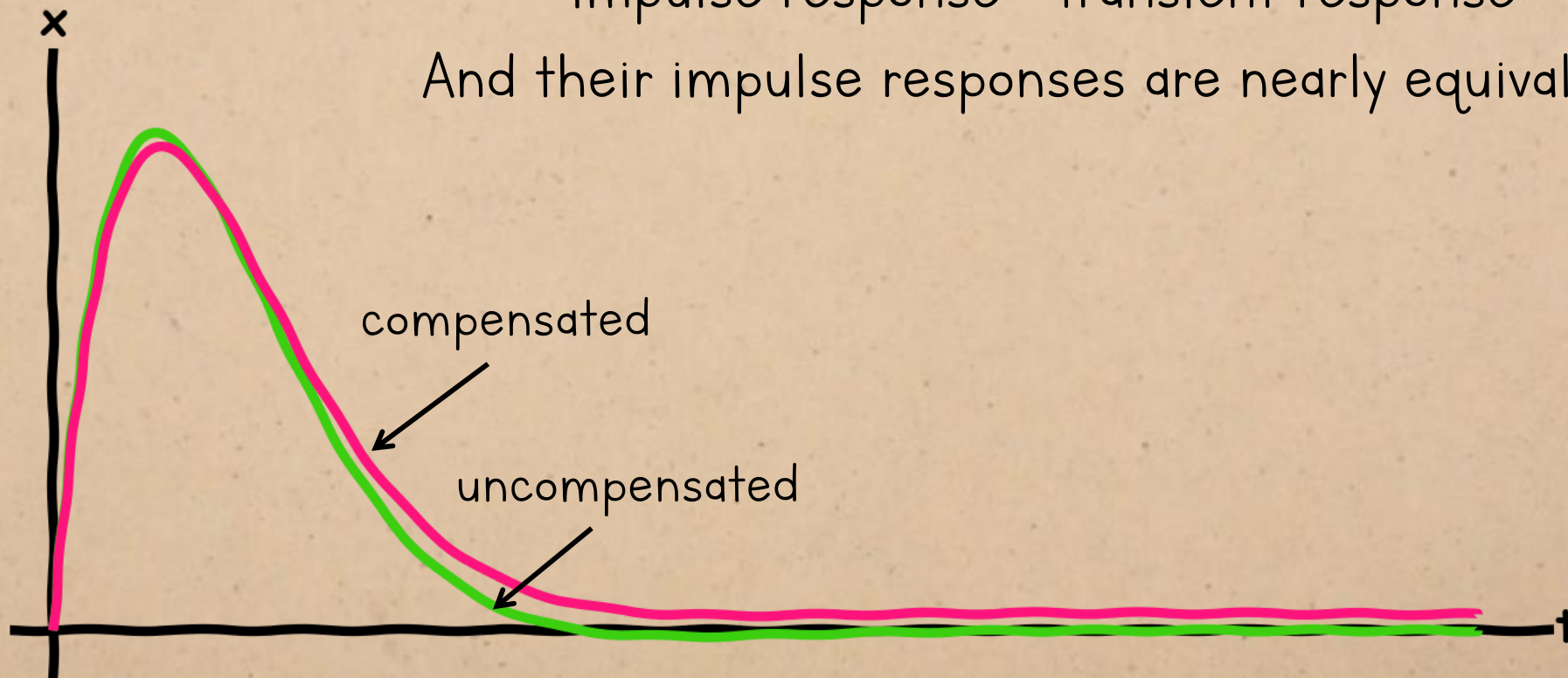
A lag compensator is useful when you have a system with an acceptable **transient response** but it doesn't have sufficient **steady-state accuracy**.

WHAT HAPPENED?

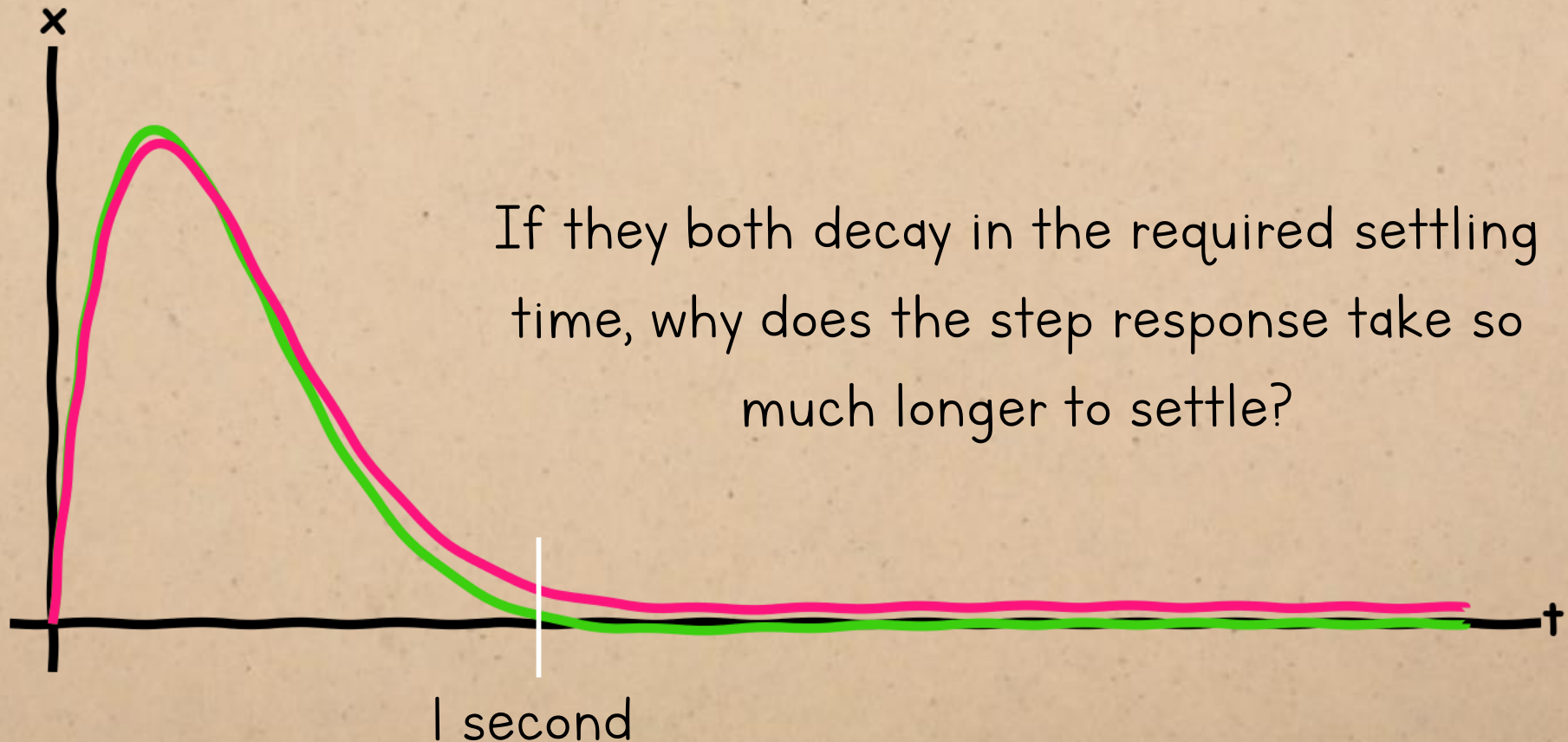
The lag compensator did preserve the transient response:

impulse response = transient response

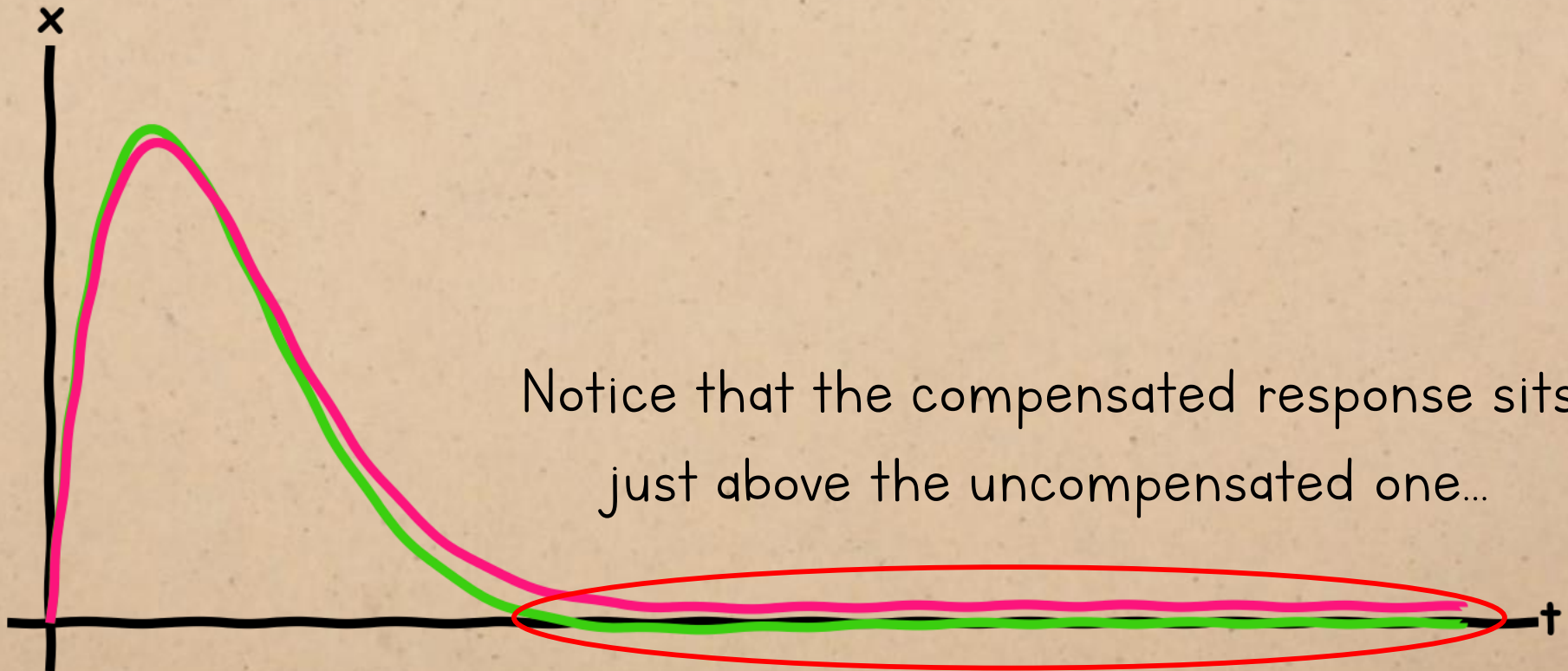
And their impulse responses are nearly equivalent.



WHAT HAPPENED?

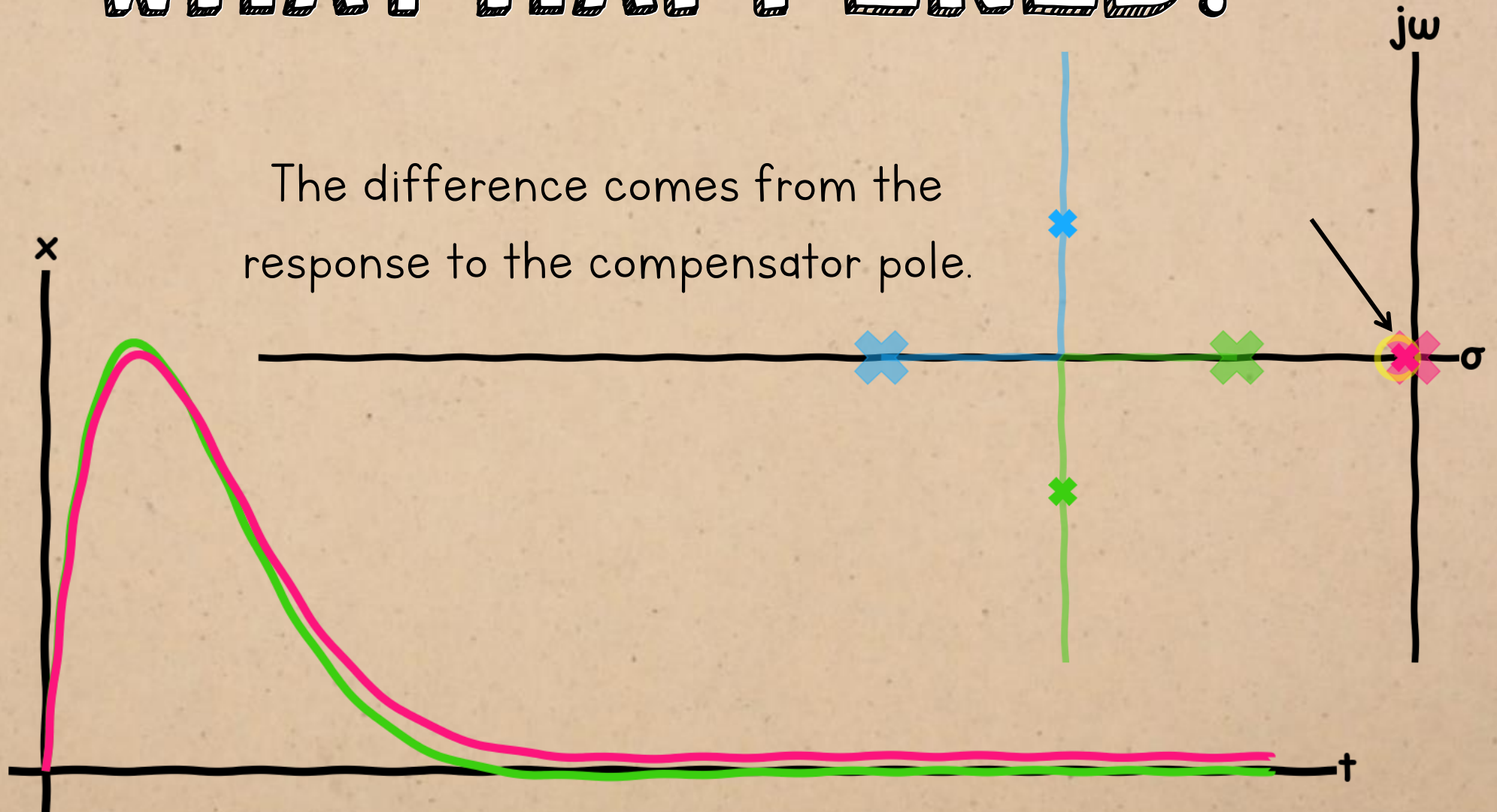


WHAT HAPPENED?



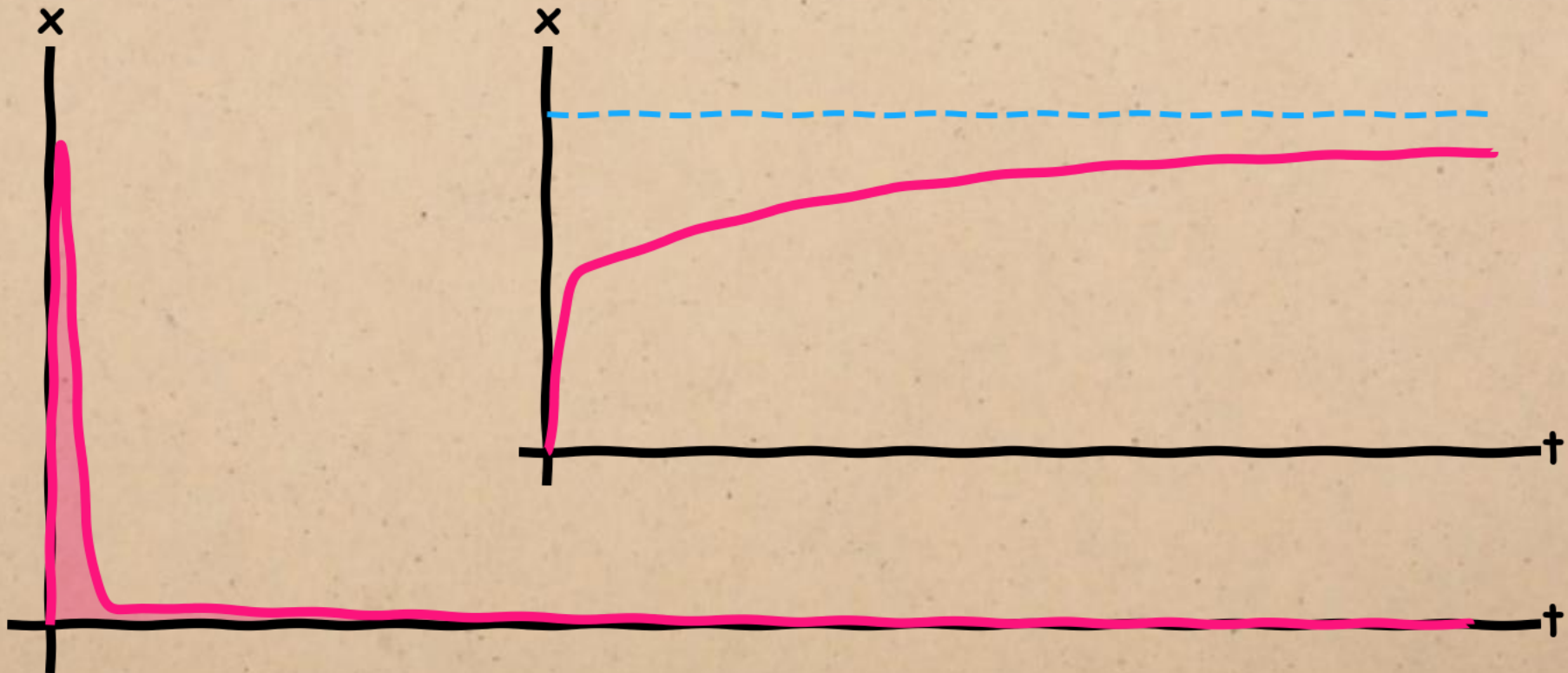
WHAT HAPPENED?

The difference comes from the response to the compensator pole.



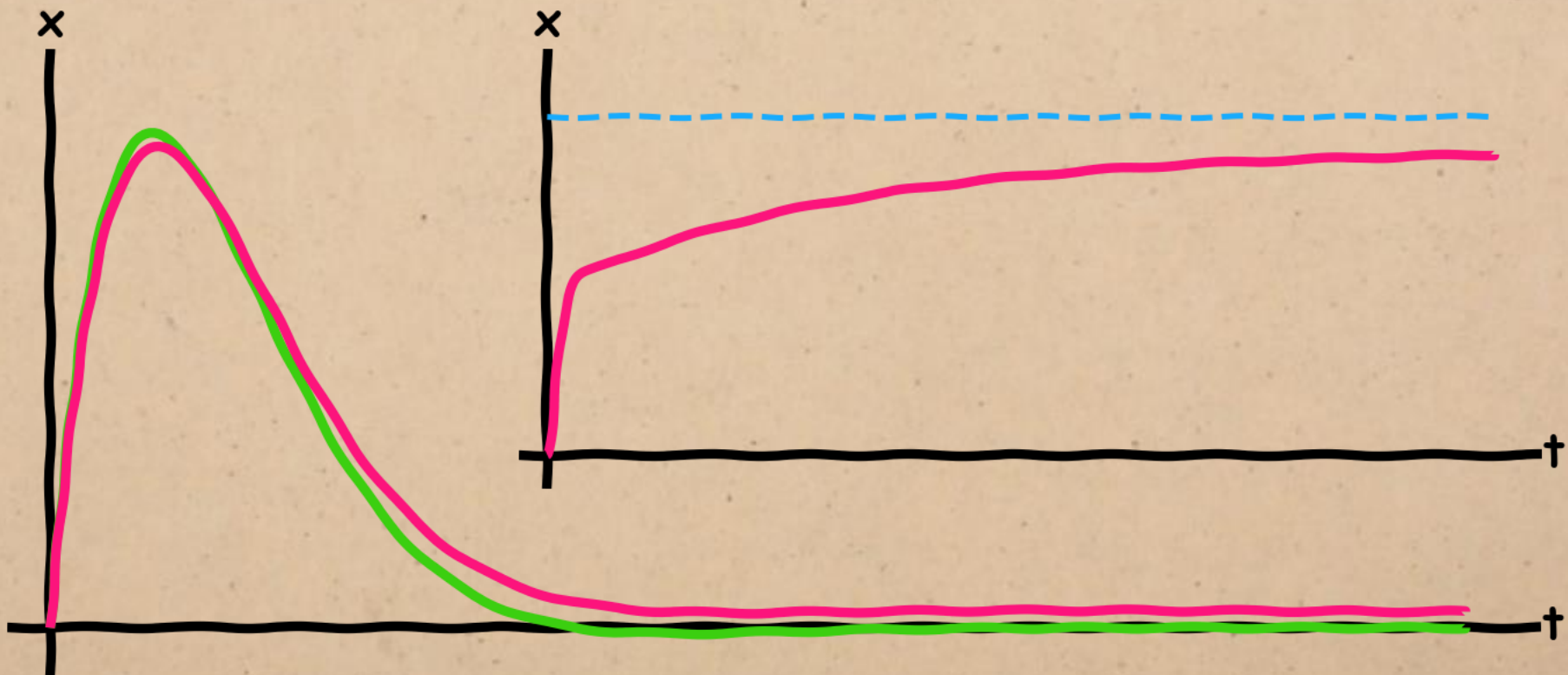
WHAT HAPPENED?

When we integrate the impulse response to get the step response, that small sliver of extra area slowly accumulates until we reach the desired position.



WHAT HAPPENED?

So the lag compensator will preserve the transient response and improve steady state accuracy, but at the cost of the settling time.



WHAT HAPPENED?

So the lag compensator will preserve the transient response and improve steady state accuracy, but at the cost of the settling time.

design a controller to meet the following requirements:

~~1. Position error $< 10\%$~~

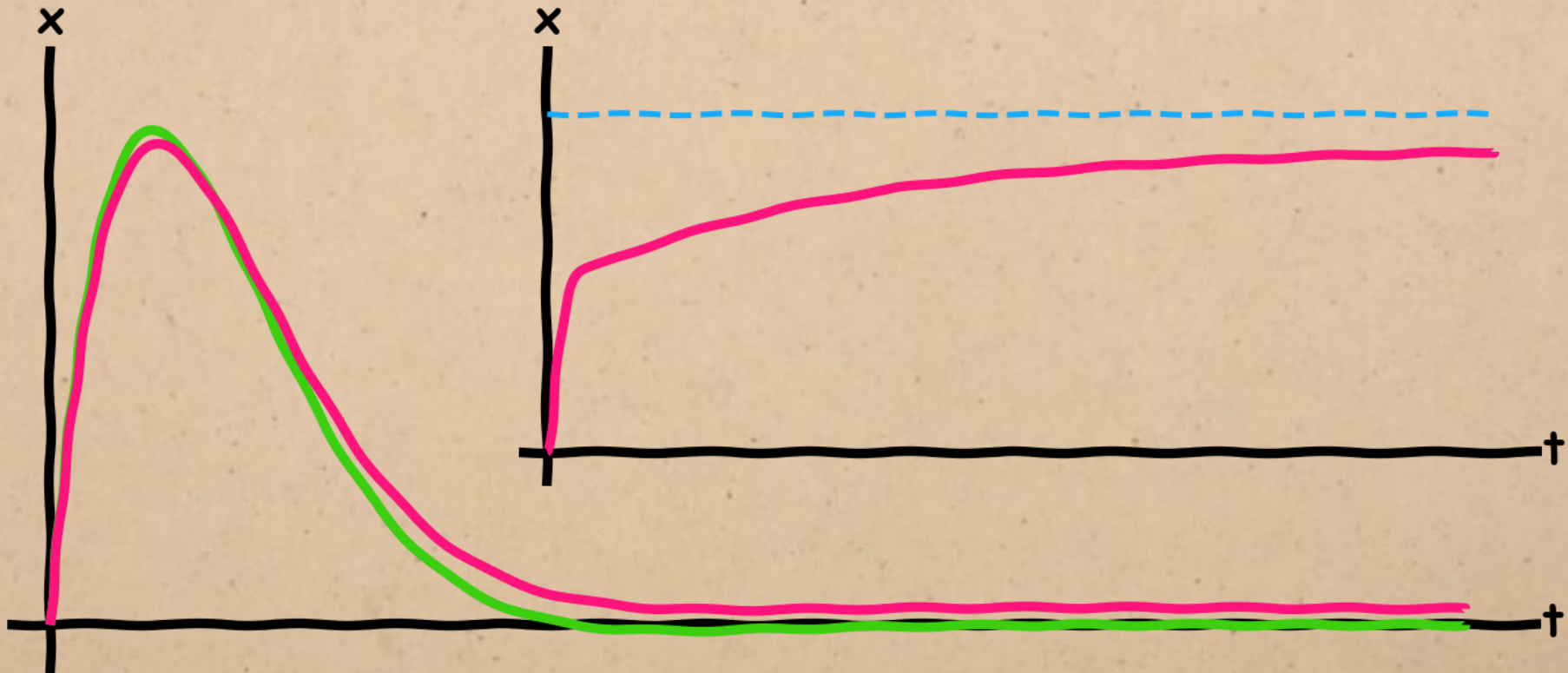
~~2. % overshoot $< 1\%$~~

3. Settling time to 2% < 1 second ?

So unfortunately, it's not really the right thing for this problem, since we can't meet this requirement...

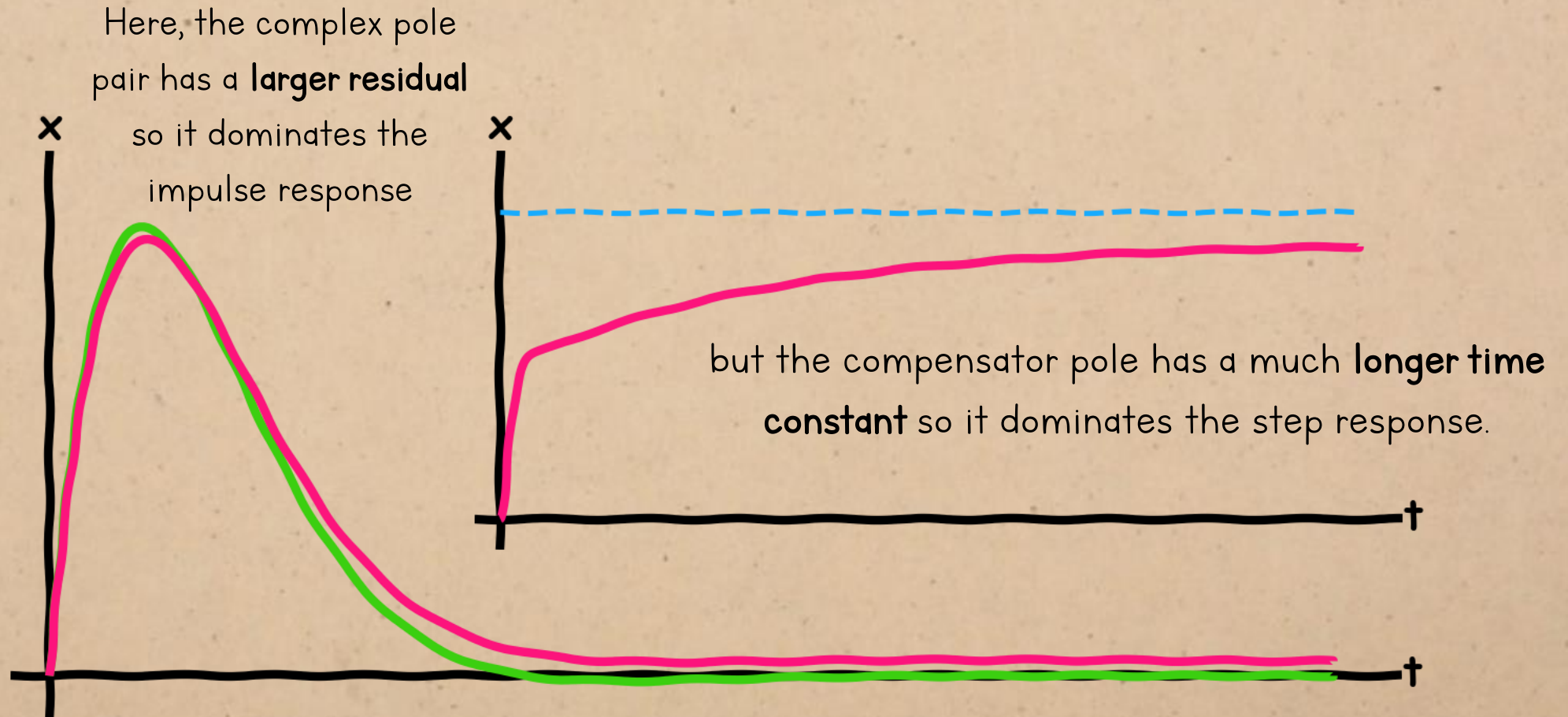
WHAT HAPPENED?

But that's ok. Some applications aren't time-critical, and the lag compensator is a good way to improve accuracy and robustness in those.



WHAT HAPPENED?

Sidenote: this illustrates how the concept of a “dominant” pole is actually context-dependent.



SUMMARY:

- Design a **lead compensator** instead. ← But there isn't one?
on the root locus
1. Identify a closed-loop pole pair [^] that gives a satisfactory transient response.

(and calculate how much gain you need to get to them using $K = \frac{1}{|G(s)|}$)

$$K \frac{s - z}{s - p}$$

2. Calculate how much more gain you need to meet the accuracy spec.

← This gives $\frac{z}{p}$

3. Place the pole and zero such that $|\theta_z - \theta_p| \leq 2^\circ$

